

Seed-Anchored Budget-Bounded Graph Rendering for Question Answering on Industry-Standard Power-Grid Information and Exchange Models

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Abstract

Large language model question answering over power-grid models must respect a fixed context budget. The artificial-intelligence contribution is a render-boundary locality prior with a precondition checkable before rendering: a deterministic, seed-anchored rendering algorithm with no method-specific tuned or learned parameters beyond the shared hop bound and context budget, preserving every predefined seed-local answer-bearing render unit when aggregate render mass $D(S)$ stays within budget B , holding for every question evaluated. The mechanism is truncation, measured without a reader: on both budget-binding networks naive rendering keeps evidence for every single-hop item but only 0.12 and 0.00 of multi-hop items; seed-anchored keeps 1.00 on both strata. A join against reader correctness shows this failure is truncation, not reader error. The engineering application is question answering over Common Information Model (CIM) models under European Network of Transmission System Operators for Electricity (ENTSO-E) Common Grid Model Exchange Standard (CGMES) conformity validation. On a fresh SmallGrid-family bank, the pre-registered primary endpoint confirms: accuracy rises from 0.450 to 0.970 ($n=100$, one-sided McNemar $p=2.2 \times 10^{-14}$), item-level confirmation within one topology family, not network-level generalization. On an independent 72,418-object topology the attribute-lookup stratum shows

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no harm, all arms at 1.000; its multi-hop stratum is retired (non-unique gold referents). Under a common pipeline, the seed-anchored standards-native arm matched or exceeded three GraphRAG frameworks’ extracted graphs at zero construction tokens, across MicroGrid, SmallGrid and SmallGrid-3.0. The endpoint is specific to one 7B reader at an 8,000-character budget; the evidence concerns budget-bounded retrieval, not general question answering.

Keywords: retrieval, graph, budget, question answering, power grid, CIM

1. Introduction

Developing scalable retrieval methods for large-language-model (LLM) assistants is a central challenge in artificial intelligence for engineering. When the knowledge source consists of detailed power-system representations encoded in the Common Information Model (CIM) [1] and exchanged via the Common Grid Model Exchange Standard (CGMES) [2], the retrieval system must select a subgraph that fits within the finite token context of an LLM while preserving all answer-relevant information. This paper treats that problem as a budget-bounded graph-rendering task and evaluates it on European Network of Transmission System Operators for Electricity (ENTSO-E) CGMES conformity-assessment configurations and an independently sourced larger grid topology; results do not establish performance on deployed operational models. The engineering motivation is grounded lookup over a utility’s own grid model rather than open-domain question answering, with intended applications such as equipment-attribute retrieval for planning and protection review, topology and containment lookup during model validation and ENTSO-E conformity assessment, operator exploration of an unfamiliar export, and comparison across model revisions. If a retrieval failure occurs under a fixed context budget, it could undermine the reliability of automated grid audits, planning tools, and operational decision support, so an assistant’s cost, data-governance, and fidelity matter as much as its raw accuracy. Four evaluation tiers are distinguished throughout: the six ENTSO-E CGMES conformity-test configurations; the RealGrid topology; synthetic scaling networks used for answer-presence evaluation only; and operational transmission systems, which are not evaluated here.

The default recipe for such assistants is graph-based retrieval-augmented generation (GraphRAG), which specialises retrieval-augmented generation [3] to a graph: build a knowledge graph over the content, retrieve a query-relevant subgraph, and let an LLM reader answer from the rendered context [4, 5].

Almost every recent power-domain GraphRAG system builds that graph by prompting an LLM to extract entities and relations from text [6], unlike earlier hand-engineered power knowledge graphs [7]. That choice is expensive and lossy: LLM extraction spans 0.03M to 1.7M tokens per network across the three frameworks we test (31,890 to 206,254 for HippoRAG; 87,962 to 754,781 for LightRAG; 235,312 to 1,703,707 for Microsoft GraphRAG), recurs on every model revision, and recovers only part of the source relational structure [8, 9], so numeric attributes migrate between similarly named components. Yet power systems already ship a curated, schema-typed relational model in the CGMES export, so we ask whether this standards-native graph can be read directly under a fixed context budget rather than reconstructed by an LLM. A prior controlled study [10] reported the puzzle we diagnose here: parsed directly from CGMES, the standards-native graph was strictly more faithful yet lost badly on the larger network under a fixed 8,000-character budget.

The failure is retrieval, not representation. We cast budget-bounded retrieval as an ordering over render units (entity descriptions and edge renderings) truncated at a budget B . The common naive order, which renders every entity description before any edge, must fail whenever a high-degree hub (a shared voltage level, region container, or busbar; degree 381 to 481 in our data) floods the budget before the answer-bearing edges are reached (Theorem 2). The failure is representation-dependent and flips with the encoding: the raw CIM graph collapses on the bus-branch SmallGrid yet survives on its node-breaker CGMES 3.0 counterpart, where the derived support graph collapses instead (Section 6.5).

A budget-safe fix, and where its novelty lies. We introduce seed-anchored retrieval, a deterministic render order that introduces no additional tuned or learned parameters and that enforces the locality prior at the render/truncation boundary through a checkable greedy-prefix precondition ($D(S) \leq B$). The guarantee covers the predefined seed-local answer-bearing render units whenever the seed-local render mass fits the budget; we do not claim that hub-induced overflow is ruled out in general. Concretely, it renders the query object’s own description and its incident edges first, then fills outward by hop distance. The retriever consumes zero LLM graph-construction tokens, introduces no learned parameters and no tuned degree threshold, and uses the same hop bound k and budget B as the naive baseline. Its methodological contribution is *where* and *how* the locality prior is enforced, at the render/truncation boundary with a provable a priori boundary rather than at the retrieval-scoring stage. In the evidence-scoped answer-presence

audit, naive rendering places the evidence-linked gold content in context for 0.56 of questions on SmallGrid and 0.50 on SmallGrid-3.0, and seed-anchored rendering for 1.00 on both; where the budget does not bind, both reach 1.00. Ego-graph expansion and hub-pruned BFS, rerun under the same test, also reach 1.00 on both budget-binding networks, so the contribution is not presence superiority over locality priors but the checkable a priori guarantee and the fact that the method introduces no additional method-specific tuned or learned parameters beyond the shared hop bound and context budget. Here, *budget-binding* is an operational label with a measured criterion: naive serialization must both reach the 8,000-character cap and actually evict evidence, the latter established by the evidence-scoped presence audit falling below 1.00. Cap hits alone do not qualify. MicroGrid illustrates why the second condition is needed: naive hits the cap on 48 of 50 items there, yet its evidence-scoped presence remains 1.00, so the cap is reached without evidence being lost and MicroGrid is not budget-binding. SmallGrid (0.56) and SmallGrid-3.0 (0.50) fall below 1.00 and are budget-binding. The label does not mean that $D(S) > B$. Nor is the problem answered by ranking the budget fill by relevance: a score-weighted fill, standard in token-budgeted context selection [11], does not protect the answer, because a high-degree hub can itself be highly relevant to the query and still flood the window and evict the seed-incident answer edges, whereas ordering the fill by hop locality renders those edges first regardless of their relevance score. What seed-anchored adds is a checkable greedy-prefix guarantee at that boundary: whenever the seed-local render mass fits the budget, every seed-local answer is preserved (an elementary answer-preservation property, not a deep theorem, Theorem 1) and the order is never worse than naive rendering in answer presence (no-harm dominance, Proposition 2), the assurance a deployer needs before swapping a renderer and the one our failed threshold heuristic lacked. It is not a new proximity score but a deployment-safe rendering rule over the standards-native CIM/CGMES graph with no LLM extraction. That failed first attempt, a degree-threshold heuristic tuned on the development network that regressed on a new one, we report as validation history (Supplementary Material, Section S1).

Evaluation with a development and held-out split. We evaluate on six ENTSO-E CGMES conformity networks spanning 233 to 5,315 objects and two CGMES versions (2.4.15 and 3.0); the CGMES 3.0 networks are the same underlying grids re-encoded in the newer version and are held out from retrieval-method, seeding, hop-bound, threshold, and scorer development,

with parser compatibility fixes for nullable 3.0 fields disclosed separately, so the evaluation tests robustness to the encoding-version change. Across twelve arms (nine graph arms, being two standards-native representations under naive rendering, the seed-anchored method, and three LLM-extracted graphs from three frameworks read under both render orders; plus three corpus-level baselines), five question strata, deterministic scoring backed by two LLM judges, and pre-registered statistics, three findings emerge. Because the templated benchmark is self-referential (83% of gold answers appear verbatim), the strongest non-self-referential evidence is the 100-question human-phrased stress test (Section 6.7, method 0.83).

1. **Seed-anchored retrieval realizes its guarantee, and the mechanism it acts on is truncation.** Measured without a reader and split by stratum, naive rendering retains the evidence-linked gold content for every single-hop item on both budget-binding networks, but for only 0.12 of multi-hop items on SmallGrid and none on SmallGrid-3.0, while seed-anchored rendering retains all of them on both strata. An item-level join against reader correctness shows the multi-hop failure is truncation rather than reader error: where the evidence survived, the reader used it. This mechanism, not any single accuracy delta, is the paper’s load-bearing result. Seed-anchored retrieval was not observed to underperform naive on any network. After Holm correction, accuracy improves significantly on three networks: MicroGrid (0.68 to 0.92, $p=7.3 \times 10^{-3}$) and the two budget-binding networks, SmallGrid (0.52 to 0.98, $p=1.2 \times 10^{-6}$) and SmallGrid-3.0 (0.42 to 0.96, $p=8.9 \times 10^{-8}$). The two budget-binding networks clear the conservative minimum detectable effect of 0.280; MicroGrid is Holm-significant but falls below it, and is not budget-binding. The other three per-network comparisons are underpowered and are treated as inconclusive, resting on a retrieval-level no-harm property under the stated precondition rather than on a statistical non-inferiority claim about end-to-end accuracy. On the larger independent RealGrid topology, whose multi-hop stratum is retired for want of unique gold referents, the surviving attribute-lookup stratum shows all three deterministic arms at 1.000, a no-harm result rather than a dominance one (Section 6.2). The pre-registered primary confirmatory endpoint is a pooled fresh-bank test drawn on the two SmallGrid encodings (Section 6.2); it is strong item-level confirmation of the mechanism within that one topology family and is not a network-level generalization claim. The human-authored probes and RealGrid

are reported separately as robustness and transfer evidence and are not folded into it.

2. **Across three GraphRAG frameworks (LightRAG, Microsoft GraphRAG, and HippoRAG), under the tested extractor, reader, and budget, standards-native seed-anchored retrieval matches or exceeds the extracted graph representations those frameworks produce, read under a common retrieval-and-rendering pipeline, at zero graph-construction tokens (extractor-conditional)** (extraction cost 0.03M to 1.7M tokens per network, which excludes build, inference, and deployment cost; Section 6.3). One framework’s native retrieval scheme, HippoRAG’s personalized PageRank, was reimplemented and evaluated; the packaged native retrievers of all three frameworks were not run. The comparison is extractor- and reader-conditional: the development-network parity held only against a frontier proprietary extractor, and swapping in an open-weight one drops LLM-extracted GraphRAG to 0.66 while anchored stays 0.98 (Section 6.4). At fixed anchored order the standards-native graph still leads on the held-out network where the budget binds (0.96 versus 0.80) and on MicroGrid, where it does not (0.92 versus 0.82), a graph-source and fidelity effect, not a rendering-order artifact.
3. **The finding generalizes beyond graphs.** Simple lexical retrieval over the deterministic corpus rendering is also strong (0.86 to 0.98 on single-hop and topology strata), with the caveat that these questions are template-generated from that corpus: what matters is a faithful standards-native representation plus budget-aware retrieval, not an LLM in the graph-construction loop.

Contributions. Our load-bearing contribution is empirical. (i) A controlled, cross-version, twelve-arm evaluation on six ENTSO-E CGMES conformity configurations, three of which carry LLM-extracted graphs and so support the extraction comparison. At zero LLM graph-construction tokens, and under the reader, extractor, budget and framework set stated in Section 5, the standards-native graph matches or exceeds extracted graphs from three extraction paradigms (LightRAG, Microsoft GraphRAG, HippoRAG; 0.03M to 1.7M tokens per network) read under a common pipeline, with one native retrieval scheme reimplemented as a check. At fixed rendering order the standards-native graph still leads, because the tested extractors lose relational and numeric fidelity. This ties to the energy-domain finding that which CGMES encoding overflows the budget flips between standard versions,

so a deployer cannot predict it (Section 6.5). (ii) The seed-anchored retriever, a deterministic render order that introduces no additional tuned or learned parameters and enforces the locality prior at the render/truncation boundary through a checkable precondition ($D(S) \leq B$) rather than at the retrieval-scoring stage, so that, unlike a relevance-ranked budget fill, a hub that is itself relevant cannot evict the seed-incident answer edges. (iii) A formal characterization that makes the truncation failure and the no-harm condition explicit and checkable: an elementary greedy-prefix answer-preservation property (Theorem 1), its no-harm dominance corollary (Proposition 2), and a quantitative naive-failure characterization (Theorem 2, Proposition 1). (iv) Evidence-bounded practical guidance, including the cost and data-governance case for reading the standard in place rather than shipping it to an external extractor, with a model-free scaling demonstration to 10^5 objects (validating answer-presence, not end-to-end reader QA accuracy) and a 100-question human-phrased robustness probe that bound the envelope of support. (v) On a second, independently sourced base topology, RealGrid, the no-harm property holds on the attribute-lookup stratum, where all three deterministic arms reach 1.000. That topology’s multi-hop stratum is retired and reported as retired: its gold labels do not identify unique referents, so it cannot support a topology-retrieval claim in either direction. In the power-system and CGMES setting we are, to our knowledge, first to analyze budget-bounded retrieval directly on ENTSO-E CGMES test configurations and first to tie a standards-native-versus-LLM-extracted graph-source result to a budget-preservation guarantee; the underlying ordering and extraction-free themes have concurrent analogues in other domains (Section 2).

Generality beyond the power-grid application. The seed-anchored render-boundary rule is a generic budget-bounded retrieval-augmented-generation preprocessing lever, not a power-grid-specific device. It reorders the retrieved subgraph so that the query object’s description and its incident edges are rendered first and further hops are added only while the cumulative render mass stays within the budget, which makes it deterministic, free of tuned or learned parameters, and equipped with a provable greedy-prefix guarantee that preserves the seed-local answer-bearing render units whenever their render mass fits the budget. The guarantee is domain-agnostic in its proof, which assumes only a graph, a render vocabulary and a budget. It is not domain-agnostic in its evidence: it is validated here only on CIM and CGMES topologies, and we make no empirical claim about other high-degree-hub retrieval settings. What is domain-specific is the render

vocabulary and the seeding rule, not the statement of the guarantee. We validate the rule on power-grid models expressed in CIM and exchanged via CGMES, where high-degree hubs and a hard context budget make the failure mode acute and a standards-native graph is available without an extraction pass; that engineering setting is the validating application, not the limit of applicability. Positioned this way, the contribution in artificial intelligence, a render-boundary locality prior with an a priori answer-preservation guarantee, leads, and the CGMES question-answering study is its validation.

2. Related work

GraphRAG and when it helps. Letting a language model build a graph and summarize its communities for global sensemaking was popularized by Microsoft’s GraphRAG [4]; follow-on systems pursued cheaper construction (LightRAG [5]) or a memory index (HippoRAG [12]). As these pipelines were benchmarked more carefully, a qualified picture emerged: on ordinary factoid questions graph-enhanced retrieval is often no better than plain retrieval [13, 14], and its advantage erodes once an agent can search [15]; the design space has been surveyed [16], with gains concentrated in relation-dense engineering settings [17, 18]. We contribute a mechanism-level account of one recurrent failure: once a context budget is fixed, the deciding factor is often not the graph but the order in which units render up to the budget boundary.

Where a locality prior is enforced. A locality prior is central to seed-anchored retrieval, and such priors are familiar in graph retrieval: HippoRAG ranks passages by personalized PageRank over a graph seeded at the query entities [12], and ego-graph or proximity-expansion retrievers score candidate nodes and edges by graph distance from the seeds. Our point of departure is the *stage* at which the prior takes effect. Those methods act at the retrieval-scoring boundary, settling *which* units enter a candidate set; but once that set is laid into a bounded context, a second and usually unstated ordering settles *which units survive truncation*, and it was there, not at scoring, that the standards-native graph failed in the prior study. Seed-anchored retrieval places the prior at this render/truncation boundary, so that with a total render order and greedy budget fill it becomes a *checkable* precondition: whenever the seed-local render mass fits the budget, $D(S) \leq B$, answer presence is guaranteed rather than merely made likely (Section 4). A proximity-scored retriever that does not govern render order carries no comparable guarantee, since a hub that clears the scoring stage can still flood the window and evict the seed-incident answer edges. The same reasoning rules out simply

ranking the budget fill by relevance, as in score-weighted truncation and token-budgeted context selection [11]: a high-degree hub can itself be highly relevant to the query and still flood the window, so a relevance-ranked fill does not protect the seed-incident answer edges, whereas a hop-locality fill renders them first regardless of their relevance score. Consistent with this, these priors and seed-anchored perform alike at reader-free answer presence; what is new is treating the render/truncation order itself as the object of a locality prior with an a priori answer-preservation precondition, layered on top of, not competing with, proximity-based retrieval.

GraphRAG and LLM evaluation in the power domain. Recent work carried these ideas into power systems: HG-RAG constructs a hierarchical, LLM-extracted graph for power-system question answering [6], GridCodex applies a RAG-driven framework to grid codes [19], and pre-LLM systems built knowledge-graph QA for the power domain [7]; on the evaluation side, ElecBench probes LLMs on power dispatch [20] and industry assessments caution about LLM-assistant reliability in operational settings [21]. None treat the engineering standard itself as the graph source and ground truth; closing that gap was the aim of a prior controlled study [10], which the present paper extends with a formal characterization, a new retrieval method, and a six-network cross-version evaluation.

CIM/CGMES as a semantic resource. Viewing the CIM as an ontology is long-standing practice [22, 23]; IEC 61970-501 fixes the CIM RDF schema [24], and ENTSO-E publishes the conformity test configurations we use [2]. Because that schema makes a CGMES export directly queryable, this line has long treated the standard as a semantic graph accessed via SPARQL rather than reconstructed by a model, so reading the standard directly is itself established practice; our contribution is not that move but tying it to budget-bounded rendering with a preservation guarantee. A parallel line of work measures how faithfully LLMs reconstruct structured knowledge from text [9, 8], and concurrent GraphRAG work in other domains likewise argues that LLM extraction is avoidable, building the index from data-derived structure for generic documents [25] or from deterministic source-code structure [26]. Our findings offer the complementary case in which a canonical structure is already present as a governed engineering standard: the right move is not to extract it more faithfully but not to extract it at all.

CGMES tooling and conformity assessment. Established power-system tooling for CGMES, the network-analysis libraries powsybl [27] and pandapower [28] and the ENTSO-E conformity-assessment workflow, assumes

full in-memory access to the parsed model or operates on small inputs, and does not ask what survives when that model must be rendered into a bounded LLM context. In the power-system and CGMES setting, and to our knowledge, this is the first work to analyze budget-bounded retrieval directly on ENTSO-E CGMES configurations and to tie that analysis to a budget-preservation guarantee, positioning the contribution against power-system graph retrieval and not only generic AI/RAG pipelines.

Rendering into a context budget. Any deployable GraphRAG system must serialize retrieved structure into a bounded context window, and overstuffing that window is known to degrade access to relevant content placed in the middle of a long context [29]; the specific failure of a high-degree entity whose neighbourhood exceeds the window and is involuntarily truncated also surfaces in graph-retrieval attack settings [30]. Our contribution is not to discover this phenomenon but to tie it quantitatively to a specific naive render order (Section 4). Budget-aware selection itself is well studied: token-budgeted context selection optimizes a relevance and redundancy objective under a budget [11]. Our answer-preservation statement is weaker and narrower than that machinery, an exact but elementary greedy-prefix property that holds only for the locality-priority order and only for answer presence, not an approximate optimality bound over a relevance objective. What we add is placing that checkable precondition at the render/truncation boundary for a standards-native graph; we do not claim a general new guarantee for GraphRAG context assembly.

3. Background

CGMES models. A CGMES export is a collection of RDF/XML profiles (equipment, topology, steady-state hypothesis, state variables) describing a network as typed objects (substations, voltage levels, lines, transformers, machines, topological nodes) joined by schema-defined associations [1, 24, 2]. Because the schema is curated and the associations explicit, the export is already a relational model of the grid before any learning component is introduced. We parse it deterministically into three artefacts. The first is a standards-native graph G_{STD} (raw CIM graph), in which every object becomes a node carrying its numeric attributes and every retained schema association becomes an edge. The second is a derived support graph G_{HYB} , in which the raw terminal-mediated associations are replaced by five derived relations: containment, electrical connectivity, node-connection, voltage, and part-of. The third is a deterministic corpus, a template rendering of the same

facts into text (for example, “AC line segment 82-96 connects topological node Baileysv to topological node Logan. It has a length of 101.782 km, a resistance of 2.82269 ohm, a reactance of 9.23472 ohm, and operates at 132 kV.”). No language model is involved in producing the ground truth, the corpus, or the questions.

The frozen QA pipeline. Every arm shares a single pipeline, following the prior controlled study [10]. Seed entities are located by matching node names against each question; the retrieved structure is rendered into text and hard-capped at $B = 8,000$ characters (stated in characters because the renderer is deterministic and tokenizer-independent); a frozen local reader, Qwen2.5-7B-Instruct in bfloat16 with greedy decoding [31], produces an answer, and a deterministic scorer checks it against a machine-checkable gold specification. Across arms only the underlying graph and its render order differ; the seeder, budget, reader, and scorer are fixed. Questions fall into five strata: S1 single-hop attribute lookups (for example, for a given transformer, what is its nominal voltage?), S2 seed-local line-to-node topology (for example, which lines connect one bus to another under the CGMES 3.0 node-breaker representation?), S3 multi-hop containment chains, S4 aggregation counts, and S5 global superlatives.

The problem. On the larger development network the prior study reported G_{STD} under the standard rendering at 0.48 overall against 0.98 for a language-model-extracted graph, even though the standards-native graph contained strictly more correct structure; the proximate signal was truncation, with 45 of 50 contexts reaching the cap. This raises the central question: is the deficit intrinsic to standards-native representations, or an artefact of rendering into a bounded context, and if the latter, can a rendering order with a provable guarantee close the gap at zero language-model graph-construction tokens?

4. Budget-bounded retrieval on CGMES power-system graphs

This section is a formal characterization of the empirical mechanism of Section 6, not a body of deep theory. The paper’s load-bearing contribution is empirical, the zero-token cost and fidelity-versus-extraction results of Section 6.3 and the encoding-dependent overflow flip of Section 6.5; the role of this section is narrower, to make the truncation failure and the no-harm condition explicit and checkable. The statements are intentionally simple, elementary greedy-prefix facts about render order under truncation rather than new mathematics. The one a practitioner needs is no-harm dominance

(Proposition 2): seed-anchored rendering is never worse than naive rendering in answer presence, so within the guaranteed regime (seed-local answers whose render mass fits the budget) adopting it cannot regress a query, the assurance needed before replacing a deployed renderer and the one our failed threshold heuristic (Supplementary Material, Section S1) lacked. Every other statement supports it: answer preservation (Theorem 1) and the naive-rendering failure mode it improves upon (Theorem 2, Proposition 1).

Let $G = (V, E)$ be the retrieval graph (undirected; each edge carries a relation label), and fix a context budget B in characters (the frozen pipeline uses $B=8,000$).

Definition 1 (Render units and orderings). *Each node-description occurrence $x = \text{desc}(v)$ and each edge-rendering occurrence $x = \text{rend}(e)$, for $e = (u, r, w)$, is a render unit identified by its source node or edge; occurrences remain distinct even if their rendered strings coincide. The quantity $\text{len}(x)$ is the unit’s rendered length including its separator. Fix injective, deterministic identifiers $\text{id}_V : V \rightarrow \mathbb{N}$ and $\text{id}_E : E \rightarrow \mathbb{N}$, and set*

$$\text{id}(\text{desc}(v)) = \text{id}_V(v), \quad \text{id}(\text{rend}(e)) = \text{id}_E(e).$$

For a query q with seed set $S = S(q) \subseteq V$ (the deterministic name-match rule shared by all arms) and hop bound k (the frozen experiments use $k = 2$), the candidate universe is

$$U_k(S) = \{\text{desc}(v) : d(v, S) \leq k\} \cup \{\text{rend}(e) : e = (u, r, w), d(u, S) \leq k, d(w, S) \leq k\},$$

where $d(\cdot, S)$ is hop distance to the nearest seed. The hop level of a unit is

$$\text{hop}(\text{desc}(v), S) = d(v, S), \quad \text{hop}(\text{rend}(e), S) = \max\{d(u, S), d(w, S)\}.$$

Thus $U_k(S)$ is exactly the set of candidate units with hop level at most k . Define the anchored tier

$$\text{tier}_S(x) = \begin{cases} 0, & x = \text{desc}(v), v \in S, \\ 1, & x = \text{rend}(e), e \text{ is incident to a vertex in } S, \\ 2d(v, S), & x = \text{desc}(v), v \notin S, \\ 2\text{hop}(x, S) + 1, & x = \text{rend}(e), e \text{ is not incident to } S. \end{cases}$$

The naive and anchored rank keys are

$$\begin{aligned}\kappa_{\text{nai}}(x) &= \begin{cases} (0, \text{id}(x)), & x \text{ is a description unit,} \\ (1, \text{id}(x)), & x \text{ is an edge-rendering unit,} \end{cases} \\ \kappa_{\text{anc}}(x) &= (\text{tier}_S(x), \text{id}(x)).\end{aligned}$$

The orders π_{nai} and π_{anc} are the increasing lexicographic orders of these keys. Injectivity of the identifiers makes both keys total orders on every finite $U_k(S)$. The context $C(\pi, B)$ is the set of units in the maximal prefix of π whose total length is at most B (the greedy hard cap used by the pipeline).

Standing assumptions. G is finite, so $U_k(S)$ is finite; every render unit has positive length; and len , measured in characters of the rendered UTF-8 text, includes the fixed single-newline separator, so a prefix's length is the sum of its units' lengths. All hold for the CGMES render units used here.

Definition 2 (Answer presence and seed-locality). *The gold answer of q is verifiable from an answer-bearing unit set (minimal with respect to set inclusion) $A(q) \subseteq U_k(S)$; for S1 attribute questions $A(q) = \{\text{desc}(s)\}$ for a seed s , and for S2 topology questions $A(q)$ is the set of answer-bearing edge-rendering units incident to the seed. Query q is retrieval-satisfied by π iff $A(q) \subseteq C(\pi, B)$. The set $A(q)$ is seed-local iff every unit in it is a seed description or a seed-incident edge rendering. Let*

$$\begin{aligned}\text{SL}(S) &= \{x \in U_k(S) : x \text{ is a seed description} \\ &\quad \text{or a seed-incident edge rendering}\}\end{aligned}$$

and $D(S) = \sum_{x \in \text{SL}(S)} \text{len}(x)$.

The analysis compares the two total orders above (Figure 1). The naive order renders all entity descriptions by node identifier and then all edges by edge identifier. The anchored order renders seed descriptions first, then every seed-incident edge, and then, for $j = 1, \dots, k$, the as-yet-unemitted j -hop descriptions followed by the as-yet-unemitted j -hop edges, with identifiers breaking ties inside each tier. In particular, the seed-local units are the first two tiers; a seed-incident edge is not emitted again in its ordinary hop tier.

Scope: the analysed stream and the implemented stream. The analysis above assigns an edge a hop level using the maximum endpoint distance, that is $\max\{d(u, S), d(w, S)\}$. The implementation instead assigns edge priority using the minimum endpoint distance, $\min\{d(u, S), d(w, S)\}$. The two orders

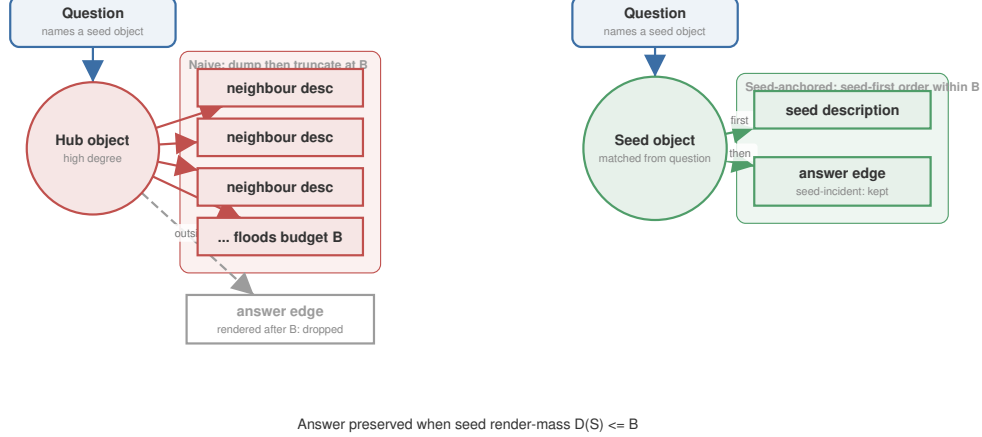


Figure 1: Rendering order under a fixed budget. Naive dump-then-truncate emits all descriptions before any edge, so a high-degree hub floods the budget and truncates the seed-incident answer edges. Seed-anchored rendering emits the seed and its incident edges first, then fills outward by hop, preserving budget-fitting seed-local answers.

therefore agree on seed-local units but can differ beyond that tier. In particular, under the implemented min-endpoint priority, an edge whose endpoint distances are $(1, 2)$ is placed in the same edge-priority tier as an edge whose endpoint distances are $(1, 1)$, even though their max-hop levels are 2 and 1 respectively. Identifier tie-breaking can therefore place an out-of-radius $(1, 2)$ edge before a remaining radius-1 $(1, 1)$ edge, so the set of max-hop radius-1 units need not form a prefix of the implemented order. Consequently, the r -local greedy-prefix argument does not apply to the shipped min-hop renderer. Theorem 1 and Proposition 2 are unaffected, because they require only that seed descriptions and seed-incident edges precede non-seed-local units, which both orders satisfy. We therefore make no r -local preservation claim beyond seed-local evidence for the evaluated implementation; the max-hop analytical extension is reported separately in Supplementary Material, Section S2. The empirical results are unaffected: every reported number comes from the implemented order.

Theorem 1 (Answer preservation). *If $A(q)$ is seed-local and $D(S) \leq B$, then $A(q) \subseteq C(\pi_{\text{anc}}, B)$.*

Proof. π_{anc} renders every unit of $\text{SL}(S)$ strictly before any non-seed-local unit. Their total length is $D(S) \leq B$, so the greedy fill admits all of $\text{SL}(S)$ into the budget prefix; $A(q) \subseteq \text{SL}(S)$ concludes. $\square$

In power-system terms, seed-anchored ordering prioritises the seed-local neighbourhood, the equipment a queried line or transformer actually connects to, so localized energy-system queries about that component stay safe under the budget. The precondition $D(S) \leq B$ is checkable a priori by summing the seed and seed-incident edge lengths before retrieval (cost $O(|S| \cdot \deg)$), and it is mild: seeds are low-degree query entities (a line, a transformer), and $D(S) \ll B$ for all S1/S2 items in the studied networks. The anchored order is not globally nondecreasing in hop when there are multiple seeds: its seed-incident tier may contain both hop-0 and hop-1 edges. Nevertheless, every seed-incident edge has hop at most 1, and after that tier the un-emitted units are ordered outward by hop.

Analytical extension, and why it is not a guarantee of the evaluated method. Under a max-hop edge-priority variant, all render units within radius r form a prefix whenever their aggregate render mass fits the budget, so an r -local preservation statement follows for the max-hop analytical order. That statement and its proof are in Supplementary Material, Section S2. The evaluated implementation instead prioritises edges by minimum endpoint distance. Since the proofs of Theorem 1 and Proposition 2 use only that every seed-local unit precedes every non-seed-local unit, a property both edge-priority variants share because the seed descriptions and seed-incident edges occupy the first two tiers under either rule, the seed-local guarantees hold for the implemented min-hop renderer as well. The present paper therefore makes no r -local preservation guarantee beyond seed-local evidence, and results for containment-chain questions are empirical. Theorem 1 and Proposition 2 are the formal guarantees for the implemented renderer; Theorem 1 is the sharper result for the seed-local tier-0/tier-1 subset, since it requires only $D(S) \leq B$ rather than the whole 1-hop mass.

Theorem 2 (Naive rendering can fail; separation). *There exist G and q with seed-local $A(q)$ and $D(S) \leq B$ such that $A(q) \not\subseteq C(\pi_{\text{nai}}, B)$.*

Proof. Proof in Supplementary Material, Section S2. □

The separation is not merely existential; the loss is quantitative in a hub’s distinct-neighbour count.

Proposition 1 (Quantitative hub flooding). *Under π_{nai} , let the entity descriptions of $U_k(S)$ have lengths summing to Δ_{desc} , rendered strictly before any edge. Let $N(h)$ be the set of distinct non-self neighbours of a hub h , and*

suppose $d(h, S) \leq k - 1$ and $m = |N(h)| \geq 1$. Then every $v \in N(h)$ lies within hop k , and

$$\Delta_{\text{desc}} \geq m \ell, \quad \ell = \min_{v \in N(h)} \text{len}(\text{desc}(v)) > 0.$$

Whenever $B < \Delta_{\text{desc}}$ (in particular, whenever $B < m \ell$), $C(\pi_{\text{nai}}, B)$ contains no edge rendering, so every edge-rendering unit in $A(q)$ is absent. If

$$L = \sum_{x \in U_k(S)} \text{len}(x), \quad T = \sum_{x \in C(\pi_{\text{nai}}, B)} \text{len}(x),$$

then $T/L \leq B/L$, and all admitted mass is description mass. If, in addition, $A(q)$ is seed-local and $D(S) \leq B$, then $A(q) \subseteq C(\pi_{\text{anc}}, B)$ by the answer-preservation theorem. Consequently, for any such query whose $A(q)$ contains at least one edge-rendering unit, naive rendering is not retrieval-satisfying while anchored rendering is retrieval-satisfying.

Proof. Proof in Supplementary Material, Section S2. $\square$

On SmallGrid ($m=381$, ℓ tens of characters, $B=8,000$) the condition $B < m \ell$ holds with room to spare, matching the near-total collapse of naive rendering to S2 accuracy 0.12 (and to 0.00 on the node-breaker SmallGrid-3.0, where the flooding hub sits on the derived support graph; Section 6.2). In power-system terms, a high-degree hub, a large substation, a shared voltage level, or a busbar, floods the budget with neighbour descriptions and truncates the incident connectivity edges, hiding the actual topology around the seed component. This flooding is the known supernode-truncation and lost-in-the-middle failure of long contexts [30, 29]; the proposition does not discover it but ties it quantitatively to the naive descriptions-first order.

Proposition 2 (No-harm dominance). *In the seed-local, $D(S) \leq B$ regime: every query retrieval-satisfied by π_{nai} is retrieval-satisfied by π_{anc} ; and when the whole universe fits ($\sum_{x \in U_k(S)} \text{len}(x) \leq B$) the two contexts contain the same unit set.*

Proof. If the universe fits, both orders render all of $U_k(S)$: identical sets. Otherwise Theorem 1 already gives $A(q) \subseteq C(\pi_{\text{anc}}, B)$ independently of π_{nai} 's outcome; containment of satisfied-query sets follows; the inclusion can be strict, as witnessed by the instances of Theorem 2. $\square$

Corollary 1. *Restricted to seed-local answers satisfying the precondition $D(S) \leq B$, seed-anchored retrieval is never worse than naive rendering in answer presence in the rendered context, so a presence regression of the kind we observed for a threshold heuristic (Supplementary Material, Section S1) is impossible by construction. This says nothing about non-seed-local queries whose answers lie outside the k -local neighbourhood (multi-hop answers, and the S3 to S5 aggregation and superlative strata), and it does not guarantee that the reader extracts an answer that is present.*

Scope. The guarantees are retrieval-level: they bound answer *presence*, a necessary condition, not whether the reader extracts it (empirical, Section 6). The formal guarantee covers S1 and S2 (Theorem 1). Results for S3 containment chains are empirical, because the implemented min-hop edge priority does not admit the radius argument. S4 aggregation answers are not single units and S5 superlatives name no seed, so neither is seed-local, and we report both as the frontier of the guarantee.

5. Method and experimental design

Our study fixes every stage of a budget-bounded retrieval pipeline and varies only the order in which render units are placed into the context window. The methodological advance is the render-boundary guarantee: a deterministic ordering that introduces no additional tuned or learned parameters and that, under the checkable budget precondition, provably preserves the seed-local answer-bearing render units whose render mass fits the budget, so that a high-degree hub cannot evict them; when that precondition fails the ordering changes which material survives truncation rather than preventing overflow. This section states the seed-anchored retrieval procedure, briefly notes the abandoned predecessor that led us to it, and specifies the networks, arms, questions, scoring, and statistics used to evaluate it.

What the benchmark measures. The primary objective is to characterize budget-bounded retrieval and ordering effects, not general-purpose question answering. The questions are template-generated deterministically from the same CGMES export that supplies the retrieval corpus, so 83% of the gold answers appear verbatim in the corpus. The rate is measured by one rule: the exact `gold_answer` string occurs in that network’s `corpus.txt`, counted over the 300 S1 and S2 items. We therefore read the benchmark as a retrieval-under-budget probe, gauging whether an answer-bearing render unit survives truncation, not whether a reader can paraphrase or reason over free text.

Algorithm 1: seed-anchored budget-bounded retrieval.*Input:* graph (V, E) , query q , budget B , hop bound k .

1. $S :=$ name-match seeds of q (shared rule, all arms).
2. Compute hop distances $d(\cdot, S)$ by breadth-first search up to k .
3. Form the unit stream: seed descriptions; seed-incident edges; 1-hop descriptions; 1-hop edges; 2-hop descriptions; 2-hop edges (ties by identifier).
4. Emit units in order; stop before the first unit that would exceed B .

Figure 2: Seed-anchored budget-bounded retrieval (Algorithm 1).

Because the benchmark is self-referential, we defer general-QA claims and treat two external checks as the load-bearing robustness evidence: a non-verbatim derived-answer subset whose gold is absent from the corpus, and a 100-question human-phrased stress test, both examined as threats in the Limitations.

5.1. Seed-anchored retrieval

Algorithm 1 (Figure 2) realizes the anchored ordering π_{anc} . A breadth-first pass labels every node with its hop distance from the query seeds, and render units are then emitted in the priority order established in Section 4 while a greedy fill consumes the budget. The procedure is deterministic and introduces no additional method-specific tuned or learned parameters beyond the shared hop bound and context budget (no learned parameters, no degree threshold, no LLM tokens), and it changes only the rendering order of the baseline retriever. The graph, the seeding rule, the hop bound, the budget, the reader, and the scorer are held identical to the naive arm, so any difference in outcome is attributable to order alone. The shared name-match seeding assumes clean identifiers; on noisy real-world inputs it can miss aliased or misspelled operator names, a robustness gap we do not address here.

The budget is a render-length budget B , measured throughout the experiments in characters and set to $B = 8,000$. We use characters rather than model tokens because the renderer is deterministic and tokenizer-independent, so the same 8,000-character cap applies uniformly across arms and readers and does not import the vocabulary of any single model into the analysis.

5.2. A refuted predecessor

Anchoring was not our first attempt: a fixed degree threshold that refused to expand through high-degree nodes lifted the collapsed stratum on the development network but then fell below the naive baseline on a previously unseen network, since no single degree cutoff separated helpful hubs from legitimate nodes that must not be blocked. We therefore adopted the ordering of Algorithm 1, which introduces no additional method-specific tuned or learned parameters beyond the shared hop bound and context budget and whose no-harm property makes regressions of this kind impossible within the guaranteed regime; the full episode is reported in the Supplementary Material, Section S1.

5.3. Networks, arms, and questions

Networks. We evaluate on six ENTSO-E CGMES Conformity Assessment test configurations from the powsybl-core mirror. Three are CGMES 2.4.15: MicroGrid (233 kept objects), MiniGrid (462), and SmallGrid (1,398). Three are their CGMES 3.0 counterparts: MicroGrid-3.0 (317), MiniGrid-3.0 (455), and the structurally distinct node-breaker SmallGrid-3.0 (5,315). The 2.4.15 exports use a bus-branch topology whereas the 3.0 exports, most sharply SmallGrid-3.0, use the more detailed node-breaker representation; the same physical grid is exchanged under either encoding, and, as we show, which representation overflows the context budget flips with that choice. The 2.4.15 trio is the development set on which the method and seeder were designed and debugged; the CGMES 3.0 trio consists of the same underlying networks re-encoded in the newer CGMES version [32] and is held out from retrieval-method, seeding, hop-bound, threshold, and scorer development. Parser compatibility fixes for nullable CGMES 3.0 fields are disclosed separately: parsing the 3.0 exports required adding `None`-guards to the CGMES parser for fields that the node-breaker encoding may legitimately omit, and those guards change the emitted corpus text. The change is recorded in the deviations register. It is a parsing-robustness fix rather than a method, threshold or hop-bound decision, but the 3.0 data did prompt a code change, so the held-out claim is scoped accordingly rather than stated absolutely.

Arms. The graph arms are G_{STD} (raw CIM graph) and G_{HYB} (hybrid derived support graph), each under the naive dump-then-truncate order; seed-anchored retrieval over G_{HYB} , our method; and G_{LLM} , a LightRAG 1.5.1 extraction (gleaning 2, 512-token chunks, local MiniLM embeddings [33]) read under the naive order. To de-confound graph source from rendering

order we add $G_{\text{LLM}+\text{ANCHORED}}$, the same extracted graph under the seed-anchored order, on the three networks with a G_{LLM} graph; and to test a second framework, Microsoft GraphRAG (graphrag 3.1.0) built with the identical open-weight extractor (gpt-oss-120b [34]), chunking (512/64), and gleaning (2), read under both orders on the same three networks; and, as a third extraction paradigm, HippoRAG 2 (2.0.0a4) [35] built via its OpenIE module under the same open-weight extractor and read under both orders on the same three networks. Three corpus arms operate over the deterministic text rendering: flat lexical RAG (tf-idf [36]), vector RAG (MiniLM cosine), and a long-context arm filling the budget with the corpus prefix. Every arm shares the same seeding rule, 8,000-character cap, frozen reader, and deterministic scorer.

Questions. Each network contributes 50 pre-registered items in strata S1 and S2 (25 single-hop attribute lookups and 25 seed-local line-to-node topology items), plus 17 to 24 extended items spanning S3 (containment chains), S4 (aggregation counts), and S5 (global superlatives). Every item is generated deterministically from the CGMES export with a machine-checkable gold specification and unique-name filtering, yielding 129 extended items in total.

5.4. Scoring and statistics

The primary scorer is deterministic, driven by a typed gold specification per item: a numeric answer matches within a relative tolerance of 1% of the gold value; an entity answer requires the gold identifier to be *asserted* as the answer, not merely to appear somewhere in the response, so a reply that restates the options and then asserts a different entity scores incorrect, while a reply that asserts the gold after a copula scores correct, and matching is insensitive to case, separators and the class prefix the item banks store; an entity-set answer requires every gold entity; and a count answer the exact integer. Abstentions are scored incorrect for every arm, and the scorer is released with the artifact. Two LLM judges corroborate it. The first is gpt-oss-120b, which is the same model used as the open-weight extractor in some evaluated arms and therefore provides only partial independence; it scores all rows of the decisive comparisons. The second is Claude Haiku 4.5 [37], a genuinely cross-family judge, which gives a secondary opinion. Agreement is reported per judge rather than pooled, and at both the per-arm and the per-stratum level, because the two levels give different ranges and a single range would obscure which is quoted. Measured against the frozen deterministic scorer, gpt-oss-120b agrees at 0.97 to 1.00 per arm and 0.94

to 1.00 per stratum, pooling to 0.987 over 2,050 judged rows. Claude Haiku agrees at 0.98 to 1.00 per arm on SmallGrid and 0.96 to 0.98 on SmallGrid-3.0, spanning 0.92 to 1.00 per stratum, pooling to 0.983 over 650 judged rows. The weakest single stratum is 0.92, on the SmallGrid-3.0 multi-hop items, and is reported rather than absorbed into a pooled range (Supplementary Material, Section S7).

The statistical plan follows the pre-registration, and we are explicit about which analyses it covers because the distinction matters for how the evidence should be read.

Primary, confirmatory. The pre-registration names a single primary test: a pooled McNemar exact test [38] of naive against seed-anchored on a freshly generated item bank spanning the two budget-binding encodings, with no multiplicity correction because there is one test. The fresh bank was not used during development and was generated after the pre-registration was frozen; it uses the already-known two budget-binding CGMES encodings, so it controls item-level exposure rather than network-level exposure. A pre-fixed sensitivity condition required the direction to hold in both encodings and both strata.

Secondary and exploratory. The six per-network McNemar tests on the original bank are *exploratory*. They were computed after the original bank had been used throughout method development, so they are post hoc with respect to that bank, and we report them Holm-corrected [39] and labelled as such rather than as a confirmatory family. Because the six networks are three base topologies re-encoded in two CGMES versions, treating them as six independent tests also understates dependence; we therefore additionally report a clustered analysis at the level of the three independent base topologies.

We report Wald 95% intervals on the exploratory and RealGrid contrasts, where an interval carries interpretive weight. The primary endpoint’s interval is the McNemar-Wald interval on the paired difference; the secondary paired contrasts against the reimplemented PPR and G_{LLM} arms use the Newcombe method-10 paired interval [40], whose endpoints differ slightly from the score-based variants of the same family. As an a priori sensitivity anchor we also report a minimum detectable effect. For a paired McNemar test the MDE is $(z_{\alpha/2} + z_{1-\beta})\sqrt{d/n}$, where d is the discordant-pair proportion, so the value depends on an assumption about d that must be stated. Under the conservative half-discordant model $d=0.5$ the MDE is 0.280 at $n=50$, $\alpha=0.05$, and power 0.80. The pre-registration recorded 0.198, which corresponds to $d=0.25$; the discordance actually observed across the six networks has mean

0.273, giving 0.207. We quote the conservative 0.280 as a design-sensitivity benchmark, not as a second significance threshold: statistical significance is decided by the test, and an effect can be significant while falling below the conservative benchmark.

The reader is Qwen2.5-7B-Instruct in bfloat16 with greedy decoding under the 8,000-character cap, run on a single NVIDIA GB10 Grace Blackwell superchip (DGX Spark, 128 GB unified memory, CUDA 13.0, PyTorch 2.12). Because decoding is greedy the runs are reproducible, and the arm tables reproduced identically on re-execution.

6. Results on ENTSO-E CGMES test configurations

We present the evidence from the retrieval-level mechanism outward: the pre-registered accuracy contrast against naive rendering, the bounded comparison against LLM-extracted GraphRAG (de-confounding graph source from render order) and its extractor sensitivity, the overflow flip, a corpus consistency check, and a model-derived human-phrased stress test, with further robustness and scope studies in the Supplementary Material. Numbers are from the deterministic scorer unless noted; the two LLM judges (gpt-oss-120b and Claude Haiku) are zero-shot with no exposure to these networks and only corroborate it. Because the questions are template-generated from the same corpus (Section 5), single-hop and topology accuracies are budget-bounded retrieval upper bounds, and the 100-question human-phrased stress test (Section 6.7, method 0.83) is the primary non-self-referential evidence, to be read before the template-generated accuracies below.

6.1. Mechanism: answer presence under the cap

Before any accuracy is measured, a reader-free audit on the development SmallGrid multi-hop stratum exposes the mechanism. Naive rendering truncated at the budget leaves the answer-bearing units in context on 3 of 25 items, all 25 contexts at the cap; seed-anchored rendering keeps the answer present on all 25 with a single context at the cap. The pattern holds across the run set (Fig. 3): naive contexts hit the cap on 32 to 50 of 50 items per network (47 to 50 on five of six), anchored on at most 2 of 50.

Three reader-free audits support the mechanism claim and are referred to throughout by short labels: E1 is a deterministic structured-query baseline over the parsed model, E2 is the evidence-scoped answer-presence audit, and E3 is the precondition audit that checks $D(S) \leq B$ per question. E2 is reported here; E1 and E3 are in Supplementary Material, Section S3.

The E2 audit tests the theorem’s empirical prediction before reader accuracy is measured. On the two budget-binding networks, naive rendering places the evidence-linked gold content in context for 0.56 of questions on SmallGrid and 0.50 on SmallGrid-3.0, and seed-anchored rendering for 1.00 on both; where the budget does not bind, both reach 1.00. These rates are pooled over the $n=50$ pre-registered S1+S2 items per network and are not multi-hop-only figures; the per-stratum split is given below and is what the mechanism argument rests on. Under the weaker substring test, which counts an equal token anywhere in the context, naive reaches 0.70 on both networks. E2 is measured against each item’s evidence path: numeric attributes are matched only in the evidence-resolved field of the emitted seed-description unit (relative tolerance 10^{-4}), entity answers only in evidence-resolved entity or edge units, and counts from the complete evidence-resolved member set; an equal token elsewhere in the context does not count. A precondition audit confirms that the seed-local render mass $D(S)$ (median 592 to 1,931 characters, maximum 3,063) satisfies $D(S) \leq B$ at $B = 8,000$ for every question on every network, so the observed anchored value of 1.00 is consistent with Theorem 1. Hop-ordered ego expansion, degree-aware hub-pruned BFS, and personalized PageRank have been rerun under this same evidence-scoped test, and we report them here. On SmallGrid and SmallGrid-3.0 respectively, over the pooled $n=50$ items, ego expansion reaches 1.00 and 1.00, hub-pruned BFS 1.00 and 1.00, and personalized PageRank [41] 0.56 and 0.50, against 1.00 and 1.00 for seed-anchored. Split by stratum, ego expansion, hub-pruned BFS and seed-anchored all reach 1.00 on S1 and on S2, whereas personalized PageRank matches naive exactly, 1.00 on S1 and 0.12 and 0.00 on S2, so its pooled parity with naive is a multi-hop failure and not an even shortfall across strata. Seed-anchored rendering is therefore not uniquely better than simple locality priors at getting the answer into the window: ego expansion and hub-pruned BFS match it on both budget-binding networks. What distinguishes seed-anchored rendering is not presence superiority but the checkable a priori guarantee and the fact that the method introduces no additional method-specific tuned or learned parameters beyond the shared hop bound and context budget: ego expansion and hub-pruned BFS reach the same presence empirically, with a tuned radius and a tuned degree threshold respectively and with no precondition that can be checked before the render is performed.

Presence is necessary, not sufficient, and separating the two bounds the claim. The E2 rates quoted above are pooled over the $n=50$ S1+S2 items per

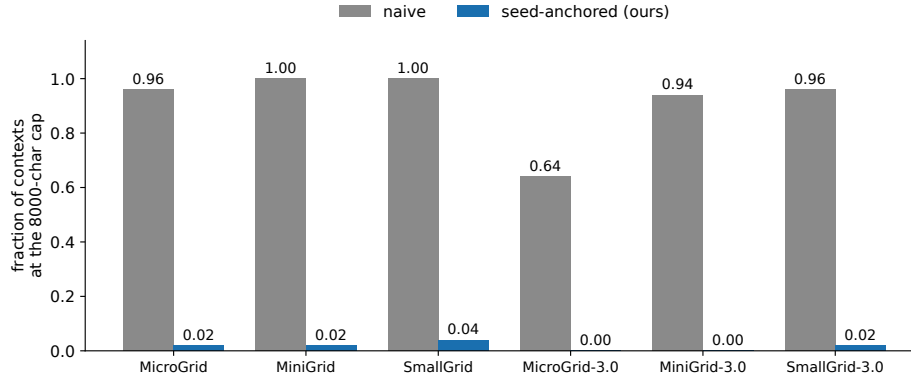


Figure 3: Fraction of contexts truncated at the 8,000-character budget for the ENTISO-E CGMES test networks under naive vs seed-anchored retrieval, per network (S1+S2). Bar values are the cap-hit counts of Table 2 expressed as fractions of $n=50$.

network; the multi-hop stratum must be read separately. Split by stratum, naive rendering places the evidence-linked gold content in context for every S1 item on both budget-binding networks (25/25), and for 0.12 of S2 items on SmallGrid (3/25) and 0.00 on SmallGrid-3.0 (0/25); seed-anchored reaches 1.00 on both strata and both networks. An item-level join of E2 presence against frozen-scorer reader correctness shows the two sets coincide exactly on S2: on SmallGrid the reader is correct on all three items where the evidence survived and on none of the twenty-two where it did not, and on SmallGrid-3.0 no S2 item retains its evidence and none is answered correctly. On this stratum the naive failure is therefore a truncation failure and not a reader failure: where the evidence was in the window the reader used it every time. The reader-side residual sits on S1 instead, where presence is 1.00 but accuracy is 0.92 on SmallGrid and 0.84 on SmallGrid-3.0, so two and four items respectively are answered incorrectly from evidence that was present. We therefore report the multi-hop failure as retrieval-only, and the single-hop residual as reader-side and outside what the theorem speaks to. The E1 structured-query analysis and E3 precondition audit are in Supplementary Material, Section S3.

6.2. Seed-anchored retrieval improves CGMES power-system QA under context budgets

The pre-registered primary endpoint. The confirmatory result is the pooled test on the fresh item bank, generated after the pre-registration was frozen and with the original bank excluded. Across the two budget-binding encodings ($n=100$), naive rendering scores 0.450 and seed-anchored 0.970, a paired

Table 1: The pre-registered confirmatory endpoint, by encoding and stratum. The pooled row is the primary test. Two readings should be kept in view: the endpoint is specific to one 7B reader at the 8,000-character budget, and gold strings occur verbatim in the corpus for 0.50 of SmallGrid items and 1.00 of SmallGrid-3.0 items, so on the latter the measurement is closer to in-context retrievability than to question answering.

Encoding	Stratum	n	naive	seed-anchored	paired difference
SmallGrid 2.4.15	S1	25	0.80	0.96	+0.16
SmallGrid 2.4.15	S2	25	0.08	1.00	+0.92
SmallGrid-3.0	S1	25	0.92	0.96	+0.04
SmallGrid-3.0	S2	25	0.00	0.96	+0.96
Pooled	both	100	0.450	0.970	+0.520

difference of +0.520 (95% CI [+0.415, +0.625]; discordant pairs $b=2$, $c=54$; one-sided McNemar exact $p=2.2 \times 10^{-14}$). Its scope should be read precisely: the fresh bank is fresh at the item level and is drawn on the two SmallGrid encodings, which are the development topology family, so it controls template and item exposure but not network-level exposure. Evidence on a topology never seen during development is the RealGrid replication below, which is deterministic-arms-only. A pre-fixed comparability check found the worst fresh-versus-original accuracy gap to be 8.0 percentage points, inside the 10-point threshold declared in advance, so the confirmatory claim carries no comparability caveat. Table 1 gives the endpoint by encoding and stratum. Anchored exceeded naive in each of the four encoding-by-stratum cells, which is the pre-fixed sensitivity condition. The cells demonstrate direction and robustness; the primary test is the pooled one, and no cell-level p -value is reported because none was pre-specified.

Exploratory per-network contrasts on the original bank. Table 2 and Fig. 4 report the six per-network contrasts. These are exploratory: the original bank was used throughout method development, so they are post hoc with respect to it. The point estimate favours seed-anchored rendering on every one of the six networks. After Holm correction, the gain is significant on three networks: MicroGrid rises from 0.68 to 0.92 ($p=7.3 \times 10^{-3}$), SmallGrid from 0.52 to 0.98 ($p=1.2 \times 10^{-6}$), and held-out SmallGrid-3.0 from 0.42 to 0.96 ($p=8.9 \times 10^{-8}$), with respective gains of +0.24, +0.46 and +0.54. The conservative design-sensitivity benchmark is 0.280; SmallGrid and SmallGrid-3.0 exceed it and MicroGrid does not, which bears on how confidently the MicroGrid effect size should be read but does not change its significance. Only SmallGrid and SmallGrid-3.0 are budget-binding; MicroGrid is significant but not budget-binding.

Because the six networks are three base topologies re-encoded in two

Table 2: Seed-anchored vs naive retrieval on six ENTSO-E CGMES test networks (S1+S2, $n=50$ per network, corrected unit-aware deterministic scoring). Items are template-generated CGMES-based QA. The naive arm here is G_{HYB} under dump-then-truncate rendering, not G_{STD} +naive, which is reported separately in Table 3. Cap hits: contexts reaching the 8000-character budget. p : McNemar exact, Holm-corrected across the six networks. The CGMES 3.0 networks are held out from retrieval-method, seeding, hop-bound, threshold and scorer development; parser compatibility fixes for nullable 3.0 fields are disclosed in Section 5. These six contrasts are exploratory, not the confirmatory endpoint.

Network	naive	anchored	cap hits (naive)	cap hits (anch.)	Holm p
MicroGrid	0.68	0.92	48/50	1/50	7.3e-03
MiniGrid	0.88	0.92	50/50	1/50	6.3e-01
SmallGrid	0.52	0.98	50/50	2/50	1.2e-06
MicroGrid-3.0	0.88	0.98	32/50	0/50	3.8e-01
MiniGrid-3.0	0.86	0.96	47/50	0/50	3.8e-01
SmallGrid-3.0	0.42	0.96	48/50	1/50	8.9e-08

CGMES versions, we also report the analysis clustered at the level of the three independent base topologies, which removes the pseudo-replication the six-test framing invites. Two of three are significant after Holm correction over three tests: MicroGrid $+0.17$ ($p=4.4 \times 10^{-4}$) and SmallGrid $+0.50$ ($p=5.3 \times 10^{-15}$); MiniGrid $+0.07$ is not significant ($p=0.065$). The defensible summary of the exploratory analysis is therefore two of three independent base topologies, not three of six networks. On the two budget-binding networks, SmallGrid multi-hop S2 accuracy rises from 0.12 to 0.96, while on SmallGrid-3.0 naive rendering answers no multi-hop item (S2 accuracy 0.00) and anchored restores S2 to 0.92. Three comparisons remain: MiniGrid ($+0.04$, $p=6.3 \times 10^{-1}$), MicroGrid-3.0 ($+0.10$, $p=3.8 \times 10^{-1}$), and MiniGrid-3.0 ($+0.10$, $p=3.8 \times 10^{-1}$). All three fall below the MDE and are underpowered, so we treat them as inconclusive rather than as wins. No network shows a significant negative difference. For the three remaining networks, end-to-end differences are inconclusive; the theorem establishes only retrieval-level answer presence under its preconditions and does not establish accuracy non-inferiority. These six per-network McNemar tests are exploratory, as set out in Section 5; the confirmatory result is the fresh-bank endpoint reported above.

An independent topology, scoped to attribute lookup. RealGrid is an ENTSO-E CGMES 2.4.15 conformity configuration of 72,418 objects on which naive rendering saturates the 8,000-character budget for every question. Its multi-hop stratum has been *retired* and is not reported. A gold-uniqueness audit found that none of its 25 S2 golds identifies a unique referent: the topological-

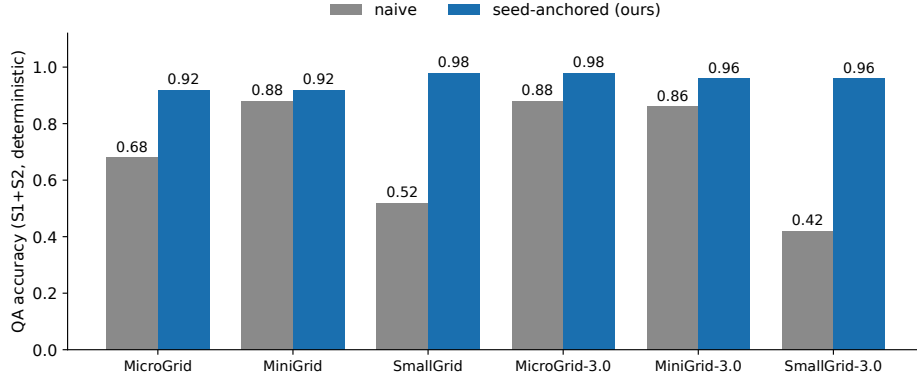


Figure 4: Seed-anchored vs naive retrieval, overall S1+S2 accuracy per network (corrected unit-aware deterministic scoring, $n=50$ each). Bar values are those of Table 2. CGMES 3.0 networks are held out.

node names used as gold labels are shared by up to 5,488 nodes, and the renderer emits names rather than machine identifiers, so no reader can assert a unique answer. The stratum does not measure what it claims to measure, so it is removed rather than narrowed, together with two subgraph-sample comparisons that rested on it.

What survives is the attribute-lookup stratum, re-derived under the frozen scorer. On S1 ($n=25$) all three deterministic arms reach 1.000: $G_{STD}+naive$, $G_{HYB}+naive$, and seed-anchored alike. RealGrid therefore supports a *no-harm* result on an independent topology, consistent with Proposition 2, and not a dominance result. LLM-extracted graphs were not rebuilt at this scale, estimated at $\sim 34M$ tokens for LightRAG alone.

All standards-native arms use the deterministic CGMES parser and spend zero LLM tokens on graph construction, whereas every arm built by LightRAG, Microsoft GraphRAG, or HippoRAG pays an LLM extraction cost, so $G_{HYB}+ANCHORED$ is a deterministic retriever rather than an LLM-enhanced one.

6.3. Against LLM-extracted GraphRAG

Table 3 and Fig. 5 set seed-anchored retrieval against LLM-extracted GraphRAG from three frameworks spanning three extraction paradigms (LightRAG keyword-relations, Microsoft GraphRAG entity-descriptions, and HippoRAG OpenIE triples), taking LightRAG first. On the development SmallGrid the two arms land in exact statistical parity (0.98 vs 0.98; paired difference 0.000, 95% CI ± 0.055), yet the extracted graph cost 594,617 tokens to build and the anchored arm none. On the held-out SmallGrid-3.0 the

open-weight gpt-oss-120b extraction (754,781 tokens) reaches 0.74 against 0.96 for anchored (+0.22, 95% CI [+0.08, +0.36], nominal $p=7.4 \times 10^{-3}$, secondary exploratory); on MicroGrid anchored again leads (0.92 vs 0.78).

Reading the LLM-extracted graph only under the naive order confounds graph source with render order, so we added **G_LLM+ANCHORED**: the seed-anchored order on the LLM-extracted graph, identical reader and scorer, on the three networks with a G_{LLM} graph (Table 3). Two findings follow. Anchored ordering is a general lever, improving the extracted graph too by +0.06 on held-out SmallGrid-3.0 (0.74 to 0.80) and +0.04 on MicroGrid (0.78 to 0.82). The lever is not uniform across frameworks: on MicroGrid, which does not bind the budget, Microsoft GraphRAG moves from 0.72 to 0.70 and HippoRAG from 0.80 to 0.78 under the same order (Table 3), so the ordering gain is non-negative for all three extraction frameworks only on the two budget-binding networks. Yet at fixed anchored order the standards-native graph still leads (0.96 vs 0.80 held-out; 0.92 vs 0.82 MicroGrid), a graph-source and fidelity effect de-confounded from render order. On development SmallGrid the seed-anchored standards-native arm and both LightRAG arms, naive and anchored, all reach 0.98, so both levers wash out there; Microsoft GraphRAG (0.40/0.46) and HippoRAG (0.80/0.86) do not, and the source effect appears only once the network binds the budget.

A second extraction framework. To test whether the against-extraction result is specific to LightRAG, we ran Microsoft GraphRAG (graphrag 3.1.0) under the identical pipeline (same open-weight extractor gpt-oss-120b, matched chunking and gleaning, CGMES-appropriate entity types, same reader, seeder, and scorer), adding its two arms to Table 3. On both budget-binding networks it scored below LightRAG and far below the zero-token anchored method (SmallGrid 0.40 naive, 0.46 anchored; SmallGrid-3.0 0.60, 0.70; against 0.98 and 0.96 for anchored), at $2.3\times$ to $2.7\times$ LightRAG’s build cost (235,312 to 1,703,707 tokens). The mechanism is an inconsistent, partly-merged entity set in which a queried component folded into a generic node has no discrete entity to anchor: S2 connectivity survives on the node-breaker 3.0 network (0.96) while S1 per-component attributes do not (0.24), the fidelity loss extraction incurs elsewhere; LightRAG under the identical seeder scored 0.98 on SmallGrid, so this is a property of the extraction, not the seeder. The full entity-merge walkthrough is in Supplementary Material, Section S7.

A third extraction paradigm. HippoRAG (OpenIE triples, the cheapest of the three at 31,890 to 206,254 tokens) confirms the render-order mecha-

nism on a third, independently extracted graph: on held-out SmallGrid-3.0 its OpenIE graph carries a flooding hub under the naive order (0.28) that seed-anchored rescues to 0.78 (+0.50), the no-harm ordering effect of Theorem 1 and Proposition 2, while the zero-token standards-native arm still leads (0.96 vs 0.78).

A framework’s own retrieval scheme, reimplemented. Reading an extracted graph under our shared renderer could understate a framework that ships its own retriever, so we reimplemented HippoRAG’s personalized-PageRank retrieval over its own extracted graph. The packaged implementation could not be run on the arm64 evaluation host, so this is a reimplementation and we label it as such. On SmallGrid it reaches 0.86 against 0.98 for seed-anchored (paired difference +0.12, 95% CI [+0.031, +0.237], McNemar exact $p=0.031$); on held-out SmallGrid-3.0 it reaches 0.78 against 0.96 (+0.18, 95% CI [+0.041, +0.319], $p=0.023$). Native PPR recovers exactly what HippoRAG’s graph reaches under our shared anchored renderer (0.86 and 0.78 on the two networks), which is direct evidence that the shared-renderer comparison is not disadvantaging the framework: the gap is in the graph, not in our rendering of it. The two remaining frameworks’ packaged retrievers were not run.

The reading is deliberately bounded: on every network, at zero LLM graph-construction tokens the standards-native *seed-anchored* arm is never behind any extracted graph under either render order. The claim is specific to that arm. The standards-native arms read under the naive order are behind on the budget-binding networks, which is the failure this paper is about: G_{HYB} under naive rendering reaches 0.52 on SmallGrid against LightRAG’s 0.98, and 0.42 on SmallGrid-3.0 against 0.74 (Table 3). So on this CIM/CGMES benchmark, conditional on the three tested frameworks, extractor, reader, and budget, LLM extraction buys no retrieval advantage. The numbers are doubly conditional: the development-network tie depended on a frontier proprietary extractor (swapping it drops G_{LLM} to 0.66, Section 6.4) and absolute accuracies are reader-dependent (Section S4); the robust claim is the ordering under fixed cost, not the level.

6.4. Extractor-controlled comparison

The development and held-out G_{LLM} graphs used different extractors (proprietary on development, open-weight gpt-oss-120b on held-out), confounding the held-out advantage with the extractor. Rebuilding the development G_{LLM}

Table 3: Full arm comparison (overall S1+S2 accuracy, deterministic retrieval; shared Qwen2.5-7B reader, seeder, and scorer), grouped by graph source across three extraction frameworks. Standards-native arms cost zero LLM graph-construction tokens; the three extraction frameworks pay the per-network build cost in each panel heading (token triples list MicroGrid / SmallGrid / SmallGrid-3.0; LightRAG and HippoRAG counts are taken from their own run logs, the Microsoft GraphRAG counts from its indexing-engine log, which is not part of the deposited artifact; Microsoft GraphRAG costs 2.3x to 2.7x LightRAG, HippoRAG is the cheapest). All three extracted graphs are read under the identical shared render pipeline, isolating graph source under matched retrieval; the anchored rows apply the seed-anchored order to the extracted graph, and with that order fixed the standards-native G_{HYB} still leads on the budget-binding networks. SmallGrid-3.0 is held out; its extracted graphs used an open-weight extractor (gpt-oss-120b). Cells are reported under the corrected unit-aware deterministic scorer, the same frozen scoring rule used in Table 2, with one exception: the MicroGrid $G_{\text{STD}}+\text{naive}$ and $G_{\text{LLM}}+\text{naive}$ entries are stored summary values for which no row-level predictions were retained, so they could not be re-derived under the frozen scorer and are carried at their stored-scorer values.

Arm	MicroGrid	SmallGrid	SmallGrid-3.0 (held-out)
<i>Standards-native (zero graph-construction tokens)</i>			
G_{STD} , naive (raw CIM)	0.84	0.48	0.94
G_{HYB} , naive (support)	0.68	0.52	0.42
G_{HYB} , anchored (ours)	0.92	0.98	0.96
<i>LightRAG extraction (87,962 / 594,617 / 754,781 tokens)</i>			
G_{LLM} , naive	0.78	0.98	0.74
G_{LLM} , anchored	0.82	0.98	0.80
<i>Microsoft GraphRAG extraction (235,312 / 1,492,123 / 1,703,707 tokens)</i>			
MS GraphRAG, naive	0.72	0.40	0.60
MS GraphRAG, anchored	0.70	0.46	0.70
<i>HippoRAG (OpenIE) extraction (31,890 / 200,785 / 206,254 tokens)</i>			
HippoRAG, naive	0.80	0.80	0.28
HippoRAG, anchored	0.78	0.86	0.78

on SmallGrid with the held-out extractor, varying only that (Table 4, single reader, scorer, seeder), drops G_{LLM} from 0.98 to 0.66 (a 691,089-token extraction, near the held-out gpt-oss level of 0.74) while the zero-token anchored arm stays 0.98. The held-out margin is thus extractor-conditional (the development-network parity required a frontier proprietary extractor at 594,617 tokens), and the de-confounding also puts seed-anchored +0.32 ahead on the development network, so that result is not a network or version artifact.

6.5. Budget overflow is representation-dependent

Table 3 hides a structural surprise. On the bus-branch SmallGrid the raw CIM graph G_{STD} collapses under naive rendering (0.48), while on the

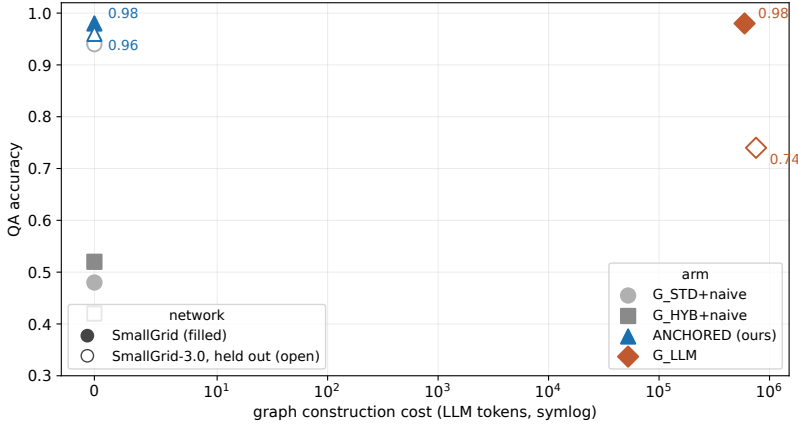


Figure 5: Accuracy vs graph-construction cost on the two budget-binding networks. All deterministic arms sit at zero tokens; LLM extraction pays 5.9×10^5 to 7.5×10^5 tokens for parity at best.

Table 4: Extractor-controlled comparison on the development SmallGrid (S1+S2, $n=50$, one reader/scorer/seeder). The two G_{LLM} rows differ only in the extractor that built the graph. Holding the extractor at the same open-weight model used for the held-out network drops G_{LLM} to the held-out level, while the zero-token anchored arm is unchanged: the held-out margin is extractor-conditional, not a network artifact.

Arm	Graph extractor	build tokens	accuracy
anchored (ours)	none (parsed CGMES)	0	0.98
G_{LLM}	Azure gpt-5.1 [42] (proprietary)	594,617	0.98
G_{LLM}	gpt-oss-120b (open-weight)	691,089	0.66

node-breaker SmallGrid-3.0 that same raw graph survives (0.94) and instead the derived support graph G_{HYB} collapses (0.42, S2 0.00), because the node-breaker encoding spreads two-hop neighbourhoods more thinly whereas the support graph’s derived shortcut edges concentrate them (Figure 6). Which representation overflows the budget is thus an artifact of the encoding, and the failing arm flips unpredictably between versions of the same network. Seed-anchored retrieval is the only arm uniformly high across every cell, the operational content of the no-harm guarantee: a deployer need not predict which graph-encoding combination will overflow.

6.6. Corpus baselines: an internal consistency check

A third family of arms moves from graphs to the deterministic *text* rendering of the standard. Flat lexical RAG is strong on S1/S2 everywhere (0.86 to 0.98, including 0.98 on SmallGrid); vector RAG degrades on the large networks (0.58 and 0.68, MiniLM [33] retrieval failing at scale); long-context

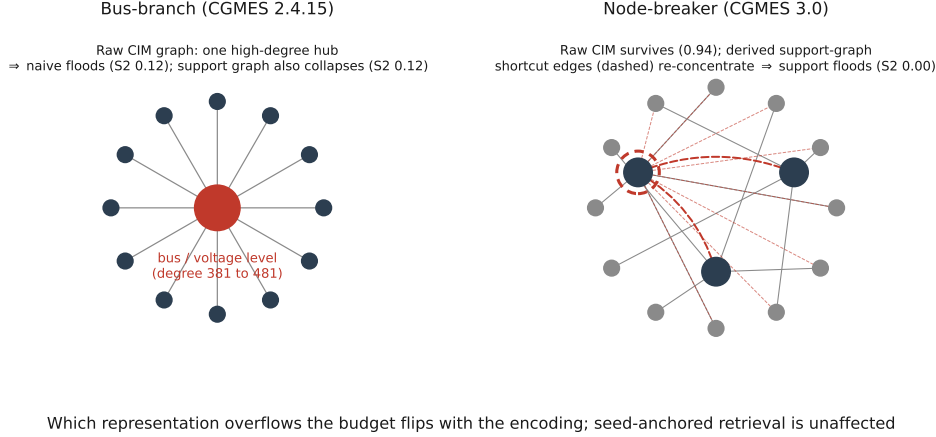


Figure 6: Why budget overflow flips with the CGMES encoding. In the bus-branch encoding (2.4.15) equipment attach directly to a high-degree bus, so the raw CIM graph carries the flooding hub and naive rendering loses the answer edges; the derived support graph is comparatively less affected (0.52 vs 0.48 overall) but collapses under naive rendering too (S2 0.12), so both graphs fail here. In the node-breaker encoding (3.0) equipment attach through breakers and connectivity nodes, which spreads raw-graph degree and lets the raw CIM graph survive (0.94), but the support graph’s derived shortcut edges (dashed) re-concentrate degree and flood instead. Seed-anchored rendering renders seed-incident edges first regardless of representation and is unaffected in either encoding.

collapses entirely there (0.00: a corpus of roughly 85,000 characters truncated at 8,000 loses most answers, the textual analogue of Theorem 2). Two caveats keep the flat-RAG result honest: 249 of the 300 S1/S2 gold answers (83%) appear verbatim, measured as exact occurrence of the `gold_answer` string in the network’s `corpus.txt`, so lexical matching enjoys a surface-form advantage that will not transfer to heterogeneous field documentation, and the corpus is itself a deterministic standards-native artifact. We therefore read flat-RAG not as evidence of generalization but as a one-directional consistency check: under the conditions stated above, a faithful standards-native representation with budget-aware retrieval matches or exceeds the extracted graphs in either graph or text form (generalization beyond the corpus is tested in Supplementary Material, Section S4).

The per-network corpus-baseline accuracies are tabulated in Supplementary Material, Section S3.

6.7. A model-derived, human-phrased stress test

To probe outside the self-referential regime we use a human-authored set of 100 operator-style questions on SmallGrid whose wording the templates never

generate, spanning eight families: attribute reads (S1), forward and reverse topology (S2), shared-node intersection (R1), unit conversion (D1), comparison (D2), two-fact arithmetic (D3), a boolean inequality (D4), and thirteen adversarial traps (false premises, name-node confusions, near-duplicate identifiers, distractors). Every gold answer is resolved deterministically from the parsed model and all 100 reconcile, so no language model touches ground truth: the set is model-derived but human-phrased, stressing robustness to non-templated adversarial wording rather than serving as an externally annotated benchmark (field-authored gold is future work). Table 5 gives accuracy by family.

The result confirms the thesis on non-templated questions and marks its boundary honestly. The proposed method (G_{HYB} with seed-anchored rendering) leads overall at 0.83, against 0.69 for the LLM-extracted arm and 0.42 for naive rendering, and the gaps fall where the theory predicts. All probe values quoted here are frozen-scorer values and are read directly from Table 5. Seed-anchored rendering reaches higher accuracy on topology (0.89 vs 0.11 naive) and shared-node intersection (1.00 vs 0.25), where two lines’ neighbourhoods share the budget and naive rendering truncates one. The LLM-extracted arm, strong on single-attribute reads (0.96) and unit conversion, loses the exact topology and values needed for intersection (0.00), comparison (0.22), and two-fact arithmetic (0.29), reinforcing that a reconstruction is not a substitute for the parsed model. On the thirteen adversarial traps the method holds 0.77 against 0.31 for naive, so the ordering advantage is not an artifact of clean phrasing. The probe is paired at $n=100$, so we test it. Exact two-sided McNemar against the proposed method gives $p=8.2 \times 10^{-10}$ versus $G_{\text{STD}}+\text{naive}$, $p=3.6 \times 10^{-8}$ versus $G_{\text{STD}}+\text{anchored}$, and $p=9.4 \times 10^{-3}$ versus the LLM-extracted arm. Families with $n \leq 4$ (R1, D1, D4) are reported descriptively as raw fractions and are not interpreted as stable family-level estimates. The honest ceiling is reader-side computation: the arithmetic and boolean families (D3, D4; D4 a single item) bound every arm, because retrieving both operands is necessary but the small reader must still compute correctly, so the residual failures sit in reader arithmetic rather than retrieval.

Held-out transfer on SmallGrid-3.0. A second 100-question human-authored set on the held-out SmallGrid-3.0 (node-breaker, 5,315 objects), run at the published budget and 2-hop retrieval with the same reader and scorer and all golds reconciled against the parsed model, shows the fidelity claim transfers to an unseen network: the proposed method leads overall (0.62) and the

Table 5: Human-phrased robustness probe on SmallGrid (100 operator-style questions; wording written by hand and not template-generated; gold resolved deterministically from the parsed CGMES model; local reader). Per-family correctness across the decisive arms, with family sizes given in the row labels. This is a single-network robustness probe, not an independent external benchmark. All cells are reported under the corrected unit-aware, assertion-aware deterministic scorer (frozen; sha256 2d9c8cfa), the same rule used for the six-network main table, and are scored against untruncated reader responses recovered by the same-item regeneration certificate, every row of which was prefix-verified against the stored text. The $G_{\text{STD}}+\text{ANCHORED}$ column reports the arm the Section 6 paired tests are computed against, on the same basis as the other three columns.

Family	G_HYB+ANCHORED	G_STD+ANCHORED	G_LLM+naive	G_STD+naive
S1: direct attribute ($n=25$)	1.00	0.92	0.96	0.84
S2: topology ($n=38$)	0.89	0.18	0.79	0.11
R1: shared-node intersection ($n=4$)	1.00	0.25	0.00	0.25
D1: unit conversion ($n=3$)	0.67	0.67	1.00	0.67
D2: value comparison ($n=9$)	0.44	0.78	0.22	0.67
D3: two-fact arithmetic ($n=7$)	0.57	0.29	0.29	0.43
D4: boolean inequality ($n=1$)	0.00	0.00	0.00	1.00
ADV: adversarial traps ($n=13$)	0.77	0.62	0.62	0.31
overall ($n=100$)	0.83	0.50	0.69	0.42

other structural-graph arms answer the questions (0.53 and 0.57), while the LLM-descriptor arm collapses to 0.16, abstaining on 84 of 100 questions, its descriptor graph summarizing rather than preserving exact attributes. Testing this probe changes how it should be read. Against the LLM-extracted arm the method is decisively ahead (exact two-sided McNemar $p=2.3 \times 10^{-12}$), and against $G_{\text{STD}}+\text{anchored}$ it is marginal ($p=0.049$); but against $G_{\text{STD}}+\text{naive}$ the difference is *not* significant (0.62 vs 0.57, $b=12$, $c=7$, $p=0.36$). On this held-out, mostly non-binding probe the method is therefore separable from the LLM-extracted arm but not from the structural naive arm, which is what the absence of a binding budget predicts. We state the consequence plainly: no held-out, non-templated comparison under a binding budget exists in this study. The held-out probe is not budget-binding, and the budget-binding comparisons are templated, so the two properties we would most want combined are never combined in a single evaluation. The SmallGrid human-probe margin against the naive arm is accordingly exploratory rather than confirmatory. This gap is partly confounded by its different open-weight gpt-oss-120b extractor (Section 6.4; Table 6, per-family detail in Supplementary Material, Section S5). We mark the boundary plainly: here SmallGrid-3.0 is mostly non-budget-binding (naive context averaging 2262 characters against the 8000-character budget, only 13 of 100 questions reaching at least 98% of it), so it does not reproduce the budget-binding dominance margin SmallGrid establishes (Table 5) and we do not present it as a second dominance result. The two networks are complementary: SmallGrid establishes the anchored-versus-naive dominance margin, and held-out SmallGrid-3.0 establishes fidelity

Table 6: Held-out human-authored probe on SmallGrid-3.0 (CGMES 3.0 node-breaker, 5,315 objects): 100 human-written questions at the published 8000-character budget and 2-hop retrieval. Overall accuracy and the budget-binding split, where binding means the G_{STD} naive rendered context reaches at least 98% of the 8000-character budget (13 of 100). Mostly non-binding here, so it tests fidelity transfer, not the dominance margin. All cells are reported under the corrected unit-aware, assertion-aware deterministic scorer (frozen; sha256 2d9c8cfa), the same rule used for the six-network main table, and are scored against untruncated reader responses recovered by the same-item regeneration certificate, every row of which was prefix-verified against the stored text. No cell is carried over from the pre-correction scorer.

Arm	Overall (100)	Binding (13)	Non-binding (87)
G_HYB+ANCHORED (published)	0.62	0.31	0.67
G_STD+naive	0.57	0.15	0.63
G_STD+ANCHORED	0.53	0.15	0.59
G_LLM+naive	0.16	0.15	0.16

transfer.

6.8. Further robustness and scope studies

Further studies, with full tables and diagnostics in the Supplementary Material, probe the result’s boundary. *Reader robustness* (Section S4): across three readers the level varies (anchored 0.98, 0.78, 0.64 on SmallGrid) but the ordering holds, anchored exceeding naive in all six reader-by-network cells and staying at or above the LLM-extracted arm on the held-out network under every reader, the development-network comparison being the sole reader-sensitive one: under gemma-3-27b the anchored arm reads 0.78 against 0.90 for the LLM-extracted arm on SmallGrid, so that cell is a deficit rather than parity, while the held-out network keeps anchored ahead under every reader. *Paraphrase robustness* (Section S4): under gpt-oss-120b paraphrases altering more than half the surface tokens while preserving identifiers (Jaccard 0.41 to 0.46), no arm collapses and the ordering holds. *Derived answers* (Section S4): on a 28-item subset whose gold sum is absent from the corpus, anchored reaches 0.83 on SmallGrid against 0.25 naive and 0.07 pooled for the LLM-extracted graph under the frozen scorer, so verbatim overlap inflates levels but not ordering. *Extended strata* (Section S5): S3 containment chains behave like seed-local questions, while S4 aggregation is largely unsolved on the large networks, with seed-anchored reaching 0.40 on SmallGrid and 0.50 on SmallGrid-3.0, and on those same large networks no retrieval arm answered an S5 superlative query, which calls for query planning. The S5 result is scoped to the large networks: on the smaller networks the stratum is not uniformly unsolved, and the anchored arm answers all three S5 items on MiniGrid-3.0. *Operational scale, model-free* (Section S6): on synthetic CGMES networks

up to 10^5 objects, anchored answer-edge presence stays 1.00 with no cap hits while naive collapses to 0.00 past the hub-flood threshold (Proposition 1); this validates answer-presence, not reader accuracy. *Cost and scorer corroboration* (Section S7): parsing G_{STD} costs zero LLM tokens whereas extraction spans 0.03M to 1.7M tokens per network and recurs with each revision, and two zero-shot cross-family judges corroborate the deterministic scorer (per stratum, gpt-oss-120b 0.94 to 1.00 and Claude Haiku 0.92 to 1.00; pooled 0.987 and 0.983 respectively).

7. Practical guidance for utilities

For a TSO placing an LLM assistant over its CIM/CGMES network models, our results in the evaluated regime (networks up to 72K objects, single 7B reader, 8,000-character budget) translate into a small set of actionable recommendations, each following from the measured behaviour of the arms and aimed at reducing the risk of missing connectivity in tasks such as protection review, reliability study, or ENTSO-E conformity assessment.

1. **Consider parsing the standard rather than reconstructing it.** The seed-anchored standards-native arm performed at least as well as the extracted graph representations produced by all three LLM-extracted GraphRAG frameworks on every network, for the tested reader, extractor, and budget, under the controlled shared-pipeline comparison that isolates graph source from each framework’s native retriever. One such retriever, HippoRAG’s personalized PageRank, was reimplemented and also stayed behind the seed-anchored arm; the packaged native retrievers were not run. LLM extraction consumed 0.03M to 1.7M tokens per network, recurs on every model revision, and left the strongest LLM-extracted arm, LightRAG, at parity with the seed-anchored arm at best and 22 points behind it at worst, with Microsoft GraphRAG further behind at 2.3x to 2.7x LightRAG’s cost (Table 3).
2. **Treat the context budget as a primary failure mode, not a tuning detail.** The errors we observed were overwhelmingly truncation of correct knowledge rather than its absence: naive rendering exhausted the budget on up to 50 of 50 queries and lost almost every multi-hop answer on the two large networks (S2 accuracy 0.12 and 0.00). Which representation overflows flips with the encoding (Section 6.5), so a clean pass on one network is no guarantee for another and each deployed encoding should be checked under its own budget.

3. **Consider seed-anchored ordering for seed-local CGMES queries** after verifying $D(S) \leq B$, seeding accuracy, and rendering completeness. It changes rendering order alone, carries an answer-preservation guarantee, adds no tuned parameters, and introduces no regression risk inside the guaranteed regime; across all six networks and both CGMES versions it stayed uniformly high.
4. **Keep the parsed CGMES model as the numeric source of truth.** LLM-extracted graphs migrate numeric attributes between similarly named components, producing confident but wrong values; grounding any numeric answer in the parsed model removes that failure class.
5. **Route aggregation and superlative queries to a query engine, not to more retrieval.** Counting questions are answered for fewer than half the items on the large networks (seed-anchored 0.40 to 0.50) and no retrieval arm answered a global-maximum question there (Supplementary Material, Section S5); no reordering of retrieved context closes that gap. These are single-line queries against the parsed model, for example SPARQL over the CIM RDF per IEC 61970-501 [24] or a call into a network-analysis library such as powsybl [27] or pandapower [28], with budget-bounded retrieval formally guaranteed for seed-local questions, while multi-hop uses remain empirical and should be separately validated. A deterministic structured-query baseline achieves perfect accuracy on regular-naming queries and degrades on node-breaker encodings where entity-name irregularity breaks template matching (Supplementary Material, Section S3). This division of labour has not been evaluated experimentally as an end-to-end hybrid system.

Reader monitoring. The retrieval-level guarantee holds independently of the reader, but the absolute accuracy level does not and varied across the readers we tested; teams should monitor the reader in production and re-validate the accuracy level when it changes, while the ordering and no-harm guarantee need no re-checking.

Deployment and data governance. The deterministic parser and retriever run entirely on the operator’s local infrastructure with no external API call, yielding reproducible, auditable retrieval that aligns with the cybersecurity and data-handling constraints utilities work under, and the tokens this avoids are a recurring per-revision cost, not a one-off. These are risk-reduction and auditability properties, not compliance claims: the method does not satisfy NERC CIP or FERC Order No. 881, and formal regulatory compliance is outside this study’s scope; “trustworthy” here means graph

fidelity, deterministic retrieval, and keeping data inside the trust boundary, not a full safety or adversarial-input assessment. The full discussion (NERC CIP, ENTSO-E, and FERC Order No. 881 context and the precise scoping of the cost claim) is in the Supplementary Material, Section S8.

8. Limitations and threats to validity

Benchmark self-referentiality (primary threat to validity). The questions are template-generated from the deterministic CGMES-derived corpus, which keeps gold answers machine-checkable and LLMs out of ground truth, but 83% of the S1/S2 gold answers appear verbatim in the corpus (Section 6.6) and the paraphrase perturbation retains identifiers, so lexical matching still succeeds; the benchmark therefore measures budget-bounded retrieval more than question-answering competence, bounding the “matches LLM-extracted graphs” claim to retrieval ordering rather than general QA accuracy. To separate faithful retrieval from verbatim recall we add a derived-answer stratum whose gold is absent as corpus spans (Supplementary Material, Section S4) and two non-templated human-phrased probes: 100 questions on SmallGrid (method 0.83, Section 6.7) and a held-out set on SmallGrid-3.0 (Section 6.7), both with the ordering and anti-extraction advantages intact. These are strong evidence rather than a full resolution: the questions span two networks, and a broader multi-network, field-authored evaluation on noisy operator documentation (missing, misspelled, or aliased identifiers) and multi-turn interaction remains future work. The guarantees are conditional on a faithfully parsed CGMES model; robustness to degraded or conversational inputs is not claimed.

Statistical power. The pre-registered confirmatory endpoint is the one-sided exact McNemar test on the fresh 100-item bank spanning the two budget-binding encodings. The six per-network contrasts discussed here are exploratory, Holm-adjusted descriptive analyses at $n = 50$ each. The MDE depends on the assumed discordant-pair proportion d : the pre-registration recorded 0.198, which assumes $d=0.25$, while the conservative half-discordant model $d=0.5$ gives 0.280 and the observed mean discordance 0.273 gives 0.207. We report the conservative 0.280 as a design-sensitivity benchmark. Three networks are significant after Holm correction: MicroGrid (+0.24, $p = 7.3 \times 10^{-3}$), SmallGrid (+0.46, $p = 1.2 \times 10^{-6}$), and SmallGrid-3.0 (+0.54, $p = 8.9 \times 10^{-8}$). Of these, SmallGrid and SmallGrid-3.0 clear the conservative MDE and are the two budget-binding networks; MicroGrid is Holm-significant but its +0.24 falls below the conservative 0.280, so we do not claim it as

an MDE-clearing effect. The other three comparisons have deltas below the MDE and are underpowered, so we treat them as inconclusive rather than as wins and do not read absence of a significant negative difference as evidence of equivalence. The answer-preservation theorem (Theorem 1) supplies a retrieval-level no-harm property under the stated precondition, a statement about answer presence; it does not establish statistical equivalence or non-inferiority of end-to-end accuracy, and the three underpowered comparisons do not establish either.

Reader dependence. The confirmatory statistics use one frozen primary reader (Qwen2.5-7B-Instruct), so absolute accuracy levels are reader-specific (anchored 0.98, 0.78, 0.64 on SmallGrid across three readers spanning two families, Supplementary Material, Section S4). Robust is the ordering, not the level: anchored is above naive in every reader-by-network cell and at or above the LLM-extracted arm on the held-out network under every reader, only the development-network anchored-versus-extraction parity being reader-sensitive. The retrieval-level mechanism is reader-independent by construction, since answer presence under the cap (Theorem 1) is a property of the rendering alone, which the reader-free audit in the artifact reproduces without any model. The main evaluation therefore rests on a single reader; the two additional reader families (Supplementary Material, Section S4) serve as a robustness check on the ordering, not as a second full evaluation, and running the confirmatory statistics on a second reader family is a planned extension.

Network diversity and behaviour at operational scale. The six networks with a full twelve-arm evaluation cover two CGMES versions but only three base topologies, so the empirical dominance claim rests on one base topology family. A fourth, independently sourced ENTSO-E base topology, RealGrid (72,418 objects, $14\times$ SmallGrid-3.0), does not lift that limitation. Its multi-hop stratum was retired after a gold-uniqueness audit found none of its 25 gold labels identifies a unique referent, with names shared by up to 5,488 nodes; two subgraph-sample comparisons that rested on that stratum were removed with it. What RealGrid supports is the attribute-lookup stratum, where all three deterministic arms reach 1.000, that is, no harm on an independent topology rather than dominance (Section 6.2). The extraction-parity claim consequently loses its RealGrid evidence and rests on the six-network comparison and the reimplemented native-PPR arm instead. LightRAG and Microsoft GraphRAG remain untested on RealGrid, disclosed future work. RealGrid does not close the gap to fully operational transmission models (10^5 to 10^6 objects); we do not claim absolute accuracies

transfer beyond it. What transfers at any scale is the seed-local retrieval-level guarantee, network-size independent since $D(S) \leq B$ (Theorem 1) bounds only the seed-local neighborhood, governed by seed degree rather than $|V|$. A model-free audit [43] confirms up to 10^5 objects (Supplementary Material, Section S6) that anchored rendering preserves the answer edge and never hits the cap while naive collapses once hub degree floods the budget, across a $32\times$ range of B . Undemonstrated is *reader* accuracy at full operational scale, requiring a licensed larger network we do not have: the main open scaling question.

Baseline scope and the LLM-extraction comparison. The LLM-extraction comparison spans three widely used builders, LightRAG, Microsoft GraphRAG (graphrag 3.1.0), and HippoRAG; other builders stay unrun, and this is disclosed as such. We read each extracted graph under the shared render pipeline with the same reader, seeder, and scorer to isolate graph source, because running each framework’s packaged native retriever would confound graph source with retrieval engine. One native scheme was nonetheless tested: HippoRAG’s personalized PageRank was reimplemented and evaluated, and it stayed behind the seed-anchored arm (Section 6.3). The packaged native retrievers themselves, including Microsoft GraphRAG’s community and local search, remain unrun. The cost argument (2.3x to 2.7x LightRAG’s tokens, all three against zero) holds regardless. Because G_{LLM} quality moves most with the extractor (Section 6.4) and the comparison is confined to these three frameworks, we rest the against-extraction claim on ordering rather than absolute margins. Whether native-retriever configurations of each framework would narrow this gap under CGMES hub topology is an open empirical question.

Guarantee scope. Theorem 1 and Proposition 2 bound answer presence, not reader extraction, and cover seed-local answers under the checkable precondition $D(S) \leq B$. S4 aggregation and S5 superlative queries lie outside the guarantee; S4 is answered for fewer than half the items on the large networks (seed-anchored 0.40 on SmallGrid and 0.50 on SmallGrid-3.0). On those two large budget-binding networks no evaluated retrieval arm answers the S5 superlative items; this is not universal across the smaller networks, where seed-anchored answers all three S5 items on MiniGrid-3.0. Reader-side arithmetic bounds accuracy even when both operands are retrieved. The precondition is a genuine, checkable gate: the model-free scaling study (Section S6) shows it flags naive’s collapse at scale, though honestly it holds for every question on every tested network, so it is never observed to reject a

query here. Three concrete cases fall outside the guarantee, and a deployer should be able to recognise them. First, a high-degree seed whose incident render mass already exceeds the budget: the greedy prefix then truncates inside the seed neighbourhood and the budget precondition $D(S) \leq B$ fails. Second, an aliased or missed seed, where a misspelled, aliased, or absent identifier prevents the seeding step from locating the query object; here the precondition may well hold and what fails instead is seeding correctness, about which the guarantee says nothing. Third, a nonlocal or global question, such as an aggregation or superlative query, whose answer lies in no seed-local neighbourhood and so falls outside the coverage of the guarantee rather than violating its precondition.

Single context budget. The primary QA runs use one budget, $B = 8,000$ characters, fixed by the prior study’s frozen pipeline. The formal results hold for arbitrary B : the precondition $D(S) \leq B$ and no-harm dominance (Proposition 2) are budget-agnostic, and a model-free budget sweep, reported in the released artifact and summarised in Supplementary Material, Section S6, confirms this across a $32\times$ range ($B < m\ell$, Proposition 1). We do not evaluate the reader accuracy-versus-budget curve or adaptive and multi-turn budgeting, left to future work.

Scoring correction. The original deterministic scorer compared a predicted number with the gold value without reading the gold `unit` field, so a numerically matching value expressed in an incompatible unit could be accepted. We re-scored all 13,402 stored predictions offline under corrected unit-aware numeric and count rules, with no new model inference. The correction changes 154 verdicts (119 numeric and 35 count). All six-network anchored accuracies are unchanged; three naive accuracies decrease by 0.02 and MicroGrid decreases by 0.04 (from 0.72 to 0.68), producing the corrected values and Holm-adjusted tests reported in Table 2.

Entity-scoring provenance, and its resolution. An audit identified an earlier entity-scoring code path that did not implement the scorer described in the method: it accepted any response containing the gold identifier, whereas the documented rule requires that identifier to be the asserted answer. We therefore re-scored the affected stored predictions under the final assertion-aware scorer. No new model inference was required.

Three values exist for the 3,503-item entity family and are distinguished here because they have been conflated elsewhere. The permissive code path scores 0.726; a strict reading that requires exact delegation scores 0.673; the frozen assertion-aware scorer, which requires the gold to be asserted but

tolerates the class prefix the banks store, scores 0.701. Every entity result in this paper and its supplement is the frozen value, 0.701. The audit table in Supplementary Material, Section S7 reports that same 0.701 over all 3,503 rows, alongside a stored-verdict column computed over the 741 of those rows that carry a recorded verdict, where the two rules disagree on 85. The correction does not change the exploratory original-bank per-network contrasts or the pre-registered fresh-bank endpoint. The entity-heavy results, including topology, shared-node intersection, RealGrid, the GraphRAG comparisons and both human probes, are therefore recomputed under the final scorer rather than preliminary. Two cross-family LLM judges agree with the frozen deterministic verdicts at 0.92 to 1.00 per stratum across both judges, pooling to 0.987 and 0.983 respectively; that agreement does not resolve either scoring-rule mismatch. The human spot-check protocol of the prior study was not repeated for the new items.

9. Conclusion

When an LLM assistant must answer questions over a standards-derived power-grid model, the graph does not have to be rebuilt by an LLM. In the evaluated comparisons, part of the apparent advantage of LLM-extracted graphs under naive rendering was a rendering-order artifact; under fixed anchored rendering the parsed graph retained an advantage in settings where the evaluated extraction lost relational or numeric fidelity. The lossiness of extraction we observe is consistent with findings reported elsewhere; the novelty here is the render-boundary mechanism and its formal precondition. Under a fixed context budget, dump-then-truncate rendering evicts answer-bearing edges as soon as a high-degree hub saturates the window, and which encoding falls into this trap changes unpredictably between representations. Seed-anchored retrieval addresses this without tuning, a rendering order with no learned parameters that carries a checkable no-harm guarantee, never worse than naive rendering in seed-local answer presence whenever the seed-local render mass fits the budget, an elementary greedy-prefix property rather than a deep theorem.

The load-bearing evidence is the truncation mechanism itself, measured without a reader. Split by stratum, naive rendering retains the evidence for every single-hop item on both budget-binding networks but for only 0.12 of multi-hop items on SmallGrid and none on SmallGrid-3.0, while seed-anchored rendering retains all of them on both strata; an item-level join against reader correctness shows the multi-hop failure is truncation

rather than reader error. That is the guarantee doing the work it was stated to do. The pre-registered primary endpoint, on a fresh bank within the SmallGrid topology family, confirms the same mechanism at item level: accuracy rises from 0.450 to 0.970 at zero LLM graph-construction tokens. It is strong item-level evidence inside one topology family and is not a network-level generalization claim; seed-anchored accuracy is also significantly higher on MicroGrid, no significant negative difference was detected, and the three remaining per-network comparisons were underpowered and remain inconclusive. The human-authored probes and the independent 72,418-object topology are reported separately as robustness and transfer evidence, and neither is folded into the confirmatory endpoint.

The claim against LLM extraction is deliberately narrow and conditional on build cost: at zero graph-construction tokens the standards-native arm is never behind the extracted graph representations produced by any of the three frameworks (LightRAG, Microsoft GraphRAG, HippoRAG; 0.03M to 1.7M build tokens per network) when those representations are read under the common retrieval-and-rendering pipeline; one framework’s native retrieval scheme was reimplemented and also stayed behind, while the packaged native retrievers were not run. Once the extractor is a realistic open-weight model the standards-native arm leads by +0.32 on the development network; any such margin is conditional on the tested extractor, reader, budget, and framework set, but the ordering under fixed cost is robust. In practice the retriever is a drop-in rendering rule for an LLM assistant or audit tool over CGMES exports at the scales evaluated (an independently sourced topology of 72,418 objects, with the model-free guarantee extending to 10^5 objects), supplying context under a fixed budget with guaranteed presence of every budget-fitting seed-local answer, an assurance that does not extend to grid control or the reader’s downstream reasoning.

Aggregation and global superlative queries remain unsolved and belong to query planning over the parsed model rather than context retrieval: detect such a query and route it to a generated SPARQL query over the CIM RDF (IEC 61970-501) or a network-analysis call, then let the reader compose the returned value; that routing has not been evaluated as an end-to-end hybrid system. RealGrid contributes an attribute-lookup no-harm result on an independent topology, where all three deterministic arms reach 1.000; its multi-hop stratum and the two subgraph-sample comparisons that rested on it were retired for lack of unique gold referents, so no RealGrid result speaks to topology retrieval or extraction parity in either direction (Section 6.2);

remaining energy-domain extensions include the full three-framework rebuild at RealGrid scale, FullGrid, and protection models, contingency analysis, and dynamic studies, where answering a query may require reasoning beyond the seed-local, k -local neighbourhood the present guarantee covers. More broadly, where a governed engineering standard exists, the evidence here suggests a data-driven assistant may do well to read it directly rather than reconstruct it with an LLM.

Data availability

The complete artifact (parsers, retrievers, QA generators with machine-checkable gold specifications, the 100-question human-authored stress test, per-item audit trails, the deviations register, and all result files with reproduction commands backing every number in this paper) is archived as a citable public deposit and companion code repository; the DOI and repository URL are withheld for double-anonymized review and will be provided on acceptance. The CGMES test configurations originate from the ENTSO-E Conformity Assessment Scheme, redistributed via the powSybl-core mirror.

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CRedit authorship contribution statement

Jayakumar Manoharan: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review and editing, Visualization. **Yamini Sehgal:** Software, Validation, Writing - review and editing.

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